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Driver Surge Maps for Supply Rebalancing with Downstream Matching

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Abstract. Problem definition: Ride-hailing platforms use driver-facing surge maps to rebalance spatial supply by offering zone-level bonuses. These bonuses can reshape supply by inducing nearby drivers to relocate and encouraging latent drivers to become dispatchable, but both responses are voluntary. The resulting supply creates operating value only when downstream matching can use it profitably, while the matching value of the induced supply does not typically inform the incentive decision. We study real-time driver surge map design when voluntary driver responses and downstream matching jointly determine the value of zone-level bonuses. **Methodology/results:** We formulate the driver surge map design problem (DSMDP) to optimize zone-level bonuses while anticipating voluntary driver responses and downstream matching. To assess how bonus adjustments affect net operating profit, we conduct a marginal analysis that identifies when small bonuses improve on the no-bonus map and how spatial smoothness links bonus decisions across zones. Using these marginal values, we develop an iterative solution approach (ISA) to sample dual sensitivities of the matching linear program across nearby matching regimes and guide real-time bonus updates. Theoretical analysis provides certificates of approximate stationarity and establishes conditions for stationary limits. Computational experiments show that ISA improves net operating profit over the best-performing feasible benchmark by up to 18.74% on the generated instances and by up to 8.43% in the New York City study. **Managerial implications:** Local shortage alone is not a sufficient basis for bonus allocation. Platforms should target responsive supply whose downstream matching value justifies the associated bonus payments. Latent-driver activation can support higher-value matches even when all demand is served, but the resulting gains must also cover additional payments to drivers who would already be matched without incentives. Results indicate diminishing returns to bonus spending and show that moderate spatial smoothness can preserve most of the gains from spatial targeting.

Key words: Ride-hailing operations; Driver incentives; Spatial rebalancing; Real-time optimization; Matching

1. Introduction

Ride-hailing platforms such as Lyft and Uber perform millions of real-time rider–driver matches every day (Garg and Nazerzadeh 2022). Yet spatial imbalances between rider demand and available drivers remain a persistent operational problem, resulting in long waits, inefficient pickups, and lost service opportunities. To improve matching success rates and mitigate spatial supply-demand imbalances, ride-hailing platforms adopt operational strategies on both the demand and supply sides. On the demand side, platforms use dynamic rider pricing to regulate realized demand, and

prior research has examined its implications (Banerjee et al. 2015, Cachon et al. 2017, Chen et al. 2024). On the supply side, platforms use spatial incentives to rebalance driver supply. Existing work studies how spatial prices and incentives reshape drivers' relocation and participation decisions and the resulting supply distribution (Afèche et al. 2023, Besbes et al. 2021, Bimpikis et al. 2019). In these models, the direction of influence runs primarily from incentives to driver responses and downstream matching outcomes, while the matching value of the induced supply does not in turn inform the incentive decision. This feedback naturally raises the question of how the platform should construct the spatial signal shown to drivers.

To influence drivers' relocation and participation decisions, one can provide spatial information, such as demand heatmaps or repositioning recommendations, to guide drivers toward particular areas without explicitly specifying a monetary reward (Haferkamp et al. 2024, Alnaggar et al. 2025). Another approach is to offer zone-level monetary incentives through a driver-facing surge map, as considered in this study. Specifically, the platform may assign zone-level bonuses based on the current supply–demand imbalances, and matched drivers who move to these zones receive the corresponding bonus. Such monetary incentive maps are widely used by ride-hailing platforms. For example, Uber displays driver surge maps in which areas with higher rider demand and limited driver supply are highlighted by stronger colors and associated with higher potential earnings (Uber 2026). Similarly, Lyft provides Bonus Zones, where drivers can unlock additional earnings by entering designated high-demand areas (Lyft 2026). Figure 1 shows examples of driver-facing maps. Surge maps can reshape supply through two voluntary-response channels: (1) attracting nearby online drivers to voluntarily reposition toward undersupplied areas, and (2) encouraging nearby offline drivers, or drivers from competing platforms, to become active on the platform.

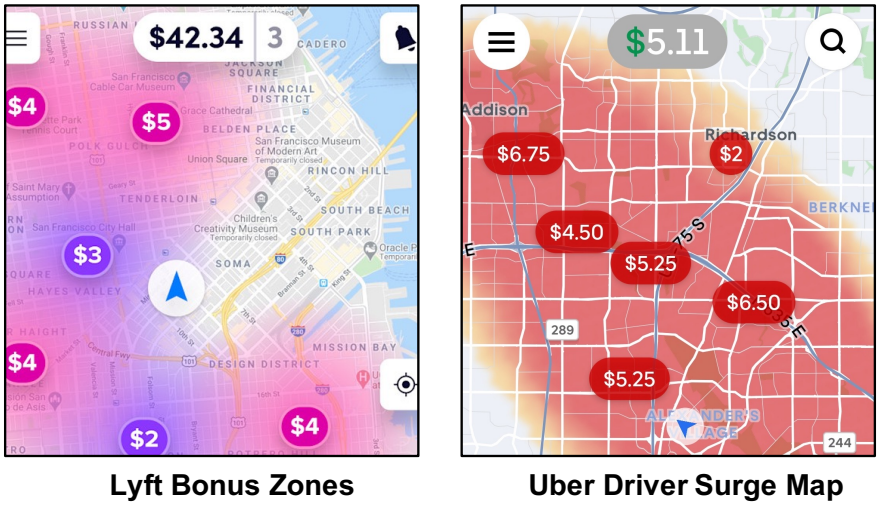


Figure 1 **Examples of supply-side spatial incentive maps on ride-hailing platforms.**

These voluntary-response channels have been studied through spatial pricing and compensation mechanisms (Bimpikis et al. 2019, Besbes et al. 2021), incentive-compatible driver payments (Garg

and Nazerzadeh 2022), and real-time positioning incentives with spatial budgets (Ong et al. 2021). Explicitly incorporating voluntary driver responses into the model can be challenging, since drivers may choose among a large set of potential relocation destinations. In real-time operations, incentive decisions are updated frequently; for example, Lyft's deployed positioning system generates fresh driver incentives every 2 minutes (Ong et al. 2021). This short decision horizon makes it natural to restrict each driver's relocation alternatives to nearby reachable locations. Consequently, driver responses over the resulting finite choice set can then be represented using discrete choice models. However, embedding such choice probabilities into the model introduces nonlinear terms, making the problem difficult to solve within a real-time operational window.

Motivated by these operational features, we study how a platform should design a zone-level driver surge map to maximize net operating profit within a short operational window. We formulate the driver surge map design problem (DSMDP), in which the platform selects zone-level bonuses while anticipating voluntary driver responses and the optimal downstream matching outcome. For simplicity, we use *latent drivers* to include drivers who are offline or serving competing platforms. To capture voluntary driver responses, we use a multinomial logit (MNL) model for dispatchable drivers' choices among nearby zones and an unavailable state, and a binary logit model for latent drivers to become available for dispatch or not within their current zones. These responses determine the expected supply available for matching, and bonuses are paid to matched drivers according to their post-response zones. To determine how downstream matching should inform bonus decisions, we analyze the matching linear program (LP) obtained by fixing the surge map.

To reflect the interaction between bonus decisions, driver responses, and downstream matching, we develop an iterative solution approach (ISA) that uses dual information from the matching problem to guide bonus adjustments. At each iteration, ISA fixes the zone-level bonuses, evaluates driver responses using the MNL and binary logit models, and solves the matching problem with the updated supply. Dual information from the matching problem then guides bonus adjustments for the next iteration. This feedback allows us to examine how marginal changes in bonuses affect driver responses and, in turn, net operating profit. Figure 2 summarizes the operational sequence. Our analysis provides finite-scale stationarity certificates and sufficient conditions for constrained Clarke stationarity of sampled limits.

We then examine the operational value of driver surge maps and show that offering bonuses improves net operating profit relative to the no-bonus map across the tested settings. The gains are greatest when available driver supply is scarce, while returns diminish as bonus spending increases. Furthermore, stronger latent-driver activation can increase net operating profit even when all demand is already served, as the additional supply supports higher-value matches. This finding highlights the role of downstream matching opportunities in determining the value of attracting

drivers. Finally, moderate limits on bonus differences between neighboring zones preserve most of the gains from spatial targeting, suggesting that effective bonus allocation does not require sharp differences across adjacent zones.

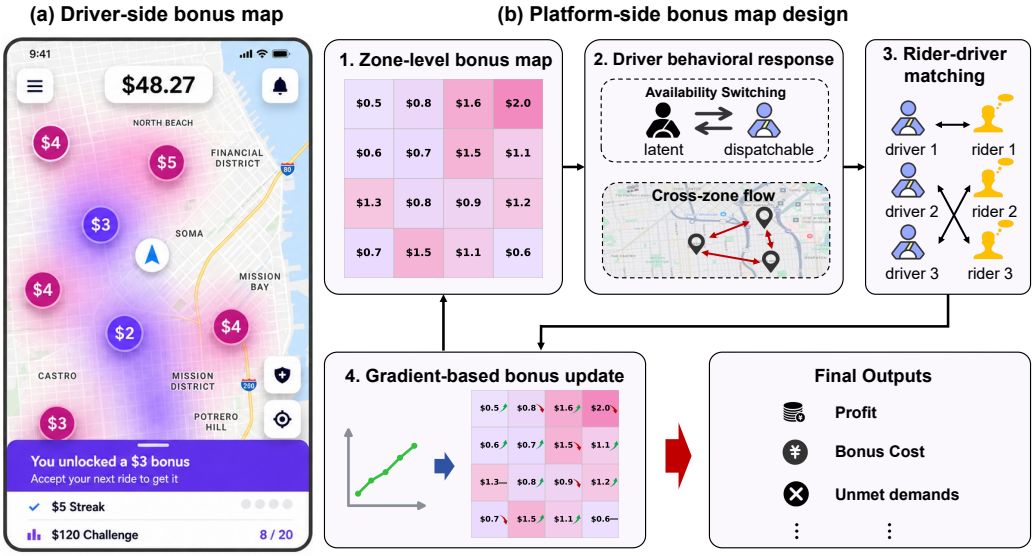


Figure 2 **Framework of the driver surge map design problem.**

We make three contributions.

- *Model formulation and structural analysis.* We formulate the DSMDP to incorporate downstream matching into spatial incentive design. We characterize the marginal value of a bonus by separating the value of responsive supply from payments on existing matches and budget opportunity cost. This yields a local screen for identifying when a small bonus improves on the no-bonus map and establishes how spatial smoothness couples marginal values across neighboring zones.
- *Real-time solution and theoretical analysis.* We develop ISA to translate the marginal-value characterization into bonus updates within the real-time decision window. The method combines matching-LP dual information with sensitivities from nearby matching regimes to address nonsmoothness in bonus optimization. We establish finite-scale stationarity certificates and sufficient conditions for constrained Clarke stationarity of sampled limits. A conic outer relaxation provides an upper bound on the optimal net operating profit.
- *Computational performance and operational insights.* ISA achieves the highest net operating profit among the evaluated methods, outperforming the best-performing feasible benchmark by up to 18.74% on the generated instances and by 2.39%–8.43% in the New York City case study. Comparisons with the no-bonus map quantify the value of intervention, while sensitivity analyses examine how value varies with budget, driver responsiveness, behavioral-parameter estimation error, and spatial smoothness.

After reviewing the literature in Section 2, we present the formulation of DSMDP in Section 3. Section 4 shows how downstream matching determines marginal bonus value, and Section 5 presents ISA and its theoretical analysis. Section 6 then reports computational performance and provides operational insights. Finally, Section 7 concludes the paper. All proofs are presented in the E-Companion.

2. Literature Review

We review three streams of literature relevant to DSMDP: surge pricing and driver bonus design, driver behavioral response and spatial relocation, and real-time matching and optimization.

2.1. Surge pricing and driver bonus design

Surge pricing and driver bonuses are central operational tools for managing supply-demand imbalance in ride-hailing platforms. A stream of studies has investigated how dynamic prices, matching decisions, and driver bonuses can be used to balance rider demand and driver supply in two-sided mobility markets. For example, Banerjee et al. (2015) developed a queueing-theoretic model for ride-sharing platforms and analyzed how pricing affects revenue and system throughput. Cachon et al. (2017) studied surge pricing on a service platform with self-scheduling capacity and showed that state-dependent prices and driver payments can improve provider utilization and market performance. From a system-level perspective, Yan et al. (2020) reviewed dynamic pricing and matching in ride-hailing platforms, examining their roles in reducing waiting times and improving marketplace reliability. Castillo et al. (2025) further showed that ride-hailing markets may suffer from "wild goose chase" failures, in which high demand, few idle drivers, and long pickup times reinforce each other and lead to substantial efficiency losses.

In ride-hailing markets, supply-demand imbalance is often spatial: some zones may face excess rider demand while nearby zones still have idle drivers. This spatial mismatch motivates a closely related stream on spatial pricing and driver bonus design. Rider-side pricing primarily regulates demand, while driver bonuses change drivers' earning opportunities and may influence their labor supply, relocation, and trip acceptance behavior (Chen 2016). Bimpikis et al. (2019) analyzed a ride-sharing network in which drivers decided whether and where to provide service under the platform's pricing and bonus policies, and Ma et al. (2022) developed spatio-temporal pricing mechanisms that accounted for demand imbalance and driver bonuses. Besbes et al. (2021) showed that localized demand shocks can induce global driver relocation patterns, while Guda and Subramanian (2019) demonstrated that surge pricing and forecast communication can serve as signals to guide worker movement across locations. These studies show how spatial pricing links local supply-demand conditions to driver responses across the network.

Another related stream has studied how driver bonus rules affect drivers' earnings and participation decisions. Garg and Nazerzadeh (2022) provided a theoretical foundation for driver-side payment mechanisms and showed that additive driver surge can improve incentive compatibility compared with multiplicative surge in dynamic settings. Bai et al. (2023) further showed that strategic providers may limit online availability to induce surge pricing, highlighting that driver-side incentives can generate strategic supply responses beyond simple participation effects.

Lyft's Personal Power Zone (PPZ) system provides a close operational analogue to our setting: it allocates real-time location-based positioning incentives under spatial budgets and computes driver-positioning decisions within seconds (Ong et al. 2021). Whereas their approach computes bonuses after determining driver-positioning allocations, our study directly optimizes a displayed driver surge map. The platform selects zone-level bonuses by linking voluntary driver relocation and changes in dispatch availability to downstream matching value net of bonus payments.

2.2. Driver behavioral response and spatial supply relocation

A key feature of ride-hailing systems is that driver supply responds behaviorally to platform bonuses. Drivers may decide whether to go online, whether to remain online, where to reposition, and which trips to accept. Lu et al. (2018) provided direct empirical evidence that surge heatmap visibility affected Uber drivers' relocation decisions and earnings, supporting the idea that map-based surge information can influence the spatial distribution of driver supply. Chen (2016) also showed that flexible workers responded to surge pricing by adjusting their labor supply decisions. Using field data, Miao et al. (2023) further showed that driver responses vary with employment type, experience, and labor-supply preferences.

Driver repositioning has also been studied from behavioral and operational perspectives. Afèche et al. (2023) examined how platform controls affect equilibrium performance in ride-hailing networks with strategic drivers. Wang et al. (2026) further characterized when spontaneous relocation mitigates imbalance and when subsidies are needed to induce platform-preferred outcomes. At the individual level, Urata et al. (2021) represented drivers' zonal search choices through a discrete choice model, while Liu et al. (2026) showed how street orders as an outside earning opportunity affect platform participation and incentive decisions. These studies account for how monetary incentives, location, and outside earning opportunities affect driver availability and relocation.

Another line of work studies heatmaps and repositioning recommendations that influence driver relocation. These tools influence drivers through spatial information or movement guidance. Alnaggar et al. (2025) and Haferkamp et al. (2024) showed that heatmap-based decision support can guide decentralized repositioning and improve spatial supply balance. In our setting, drivers respond to monetary zone-level bonuses through both relocation and changes in dispatch availability.

To represent these voluntary responses, we use an MNL model for dispatchable drivers' choices among reachable zones and an unavailable state, and a binary logit model for latent-driver activation within the current zone. These responses determine the expected post-response supply available for downstream matching.

2.3. Real-time matching, dispatching, and optimization methods

Real-time rider-driver matching and order dispatching form another major stream of ride-hailing research. Matching determines how available drivers are assigned to rider requests and directly affects passenger waiting time, driver utilization, platform revenue, and future supply distribution. Özkan and Ward (2020) studied dynamic matching for real-time ride sharing and proposed policies based on continuous linear programs that accounted for spatial heterogeneity, waiting constraints, and time-varying parameters. Yan et al. (2020) reviewed matching and pricing algorithms in ride-hailing platforms and emphasized that these decisions are often implemented at industrial scale under stringent real-time requirements.

Several studies developed scalable dispatching algorithms for industrial ride-hailing platforms. Xu et al. (2018) proposed a learning-and-planning framework that incorporates predicted future value into real-time matching, while Qin et al. (2020) described the deployment of sequential order-dispatching methods at DiDi. These studies show that real-time dispatching requires scalable algorithms that account for both immediate matching performance and future system impacts.

Another related stream considered vehicle repositioning and rebalancing. Braverman et al. (2019) provided structural insights into empty-car routing, whereas Tuncel et al. (2023) and Gao et al. (2024) integrated repositioning with matching decisions. This research examines how vehicle movements and assignments affect service performance. In DSMDP, bonuses influence voluntary driver movements and dispatch availability before the resulting supply enters the matching system.

Our study examines how downstream matching should inform zone-level bonus decisions when driver supply responds voluntarily to incentives. For a fixed driver surge map, the matching LP evaluates the post-response supply while accounting for bonus payments and the available budget. Its dual information characterizes the marginal value of zone-level bonuses and guides the feasible bonus updates in ISA.

3. Problem Description and Formulation

This section introduces the DSMDP and develops its mathematical formulation. We first describe the operational setting, then model driver responses to posted bonuses and incorporate the resulting supply into the platform's optimization problem.

3.1. Problem Description

We consider a ride-hailing system operating over discrete real-time decision periods. At the beginning of each period, the platform observes rider demand and driver availability and posts a driver surge map, to which drivers respond through voluntary relocation and changes in dispatch availability. The downstream matching system then assigns available drivers to rider requests, and bonuses are paid to matched drivers based on their post-response zones. In practice, this sequence is repeated frequently, e.g., every 2 minutes. For simplicity, we suppress the period index in the remainder of this section.

The service region is partitioned into a finite set of zones denoted by $\mathcal{Z}$, with cardinality $|\mathcal{Z}|$. Each zone serves as the basic spatial unit for displaying driver-facing bonuses and conducting real-time matching operations. In practice, platforms can define these units using a fine-grained geospatial grid, such as the cells used in Lyft's driver-positioning system (Ong et al. 2021).

The current system state is characterized by zone-level rider demand and driver supply. Specifically, d_j denotes the number of ride requests awaiting service in zone $j \in \mathcal{Z}$, while s_i^{On} and s_i^{Off} denote the dispatchable and latent driver supplies in zone $i \in \mathcal{Z}$, respectively. Dispatchable drivers are immediately available for matching, whereas latent drivers may view the surge map and become available in response to incentives. We assume that their approximate locations are available from recent application activity or location signals. Since driver bonuses can induce short-term relocation, let $\mathcal{Z}_i \subseteq \mathcal{Z}$ denote the set of zones that a driver initially located in zone i can reach within one decision period. We assume $i \in \mathcal{Z}_i$ and a symmetric reachability relation, under which $j \in \mathcal{Z}_i$ if and only if $i \in \mathcal{Z}_j$, as satisfied by the grid neighborhoods used in the experiments. This neighborhood captures the spatial scope over which the platform can reshape driver supply in real time.

Given the observed system state, the platform may regulate the spatial distribution of driver supply by posting zone-level monetary incentives. To this end, it posts a nonnegative driver-side *surge bonus* $b_i \geq 0$ for each zone $i \in \mathcal{Z}$. Following the *additive surge* structure described by Garg and Nazerzadeh (2022), the bonus is attached to the service zone and is independent of the travel distance or travel time of a particular trip. We collect these zone-level bonuses in the vector $\boldsymbol{b} = \{b_i\}_{i \in \mathcal{Z}}$, which represents the driver surge map displayed by the platform.

Designing the driver surge map requires the platform to anticipate how the offered bonuses will reshape driver supply. Specifically, dispatchable drivers may relocate across nearby zones or become unavailable for dispatch (e.g., switch to a different platform), whereas latent drivers may become available for service. These behavioral responses alter the spatial supply distribution available to the matching system and therefore the operational value of the driver surge map. We represent the downstream matching outcome by the matching vector $\boldsymbol{x} = \{x_{ij}\}_{i,j \in \mathcal{Z}}$, where x_{ij} denotes the number of available drivers in zone i assigned to ride requests in zone j during the current period.

The platform selects the driver surge map $\boldsymbol{b}$ to maximize net operating profit while anticipating the optimal downstream matching outcome $\boldsymbol{x}$. For a match between a driver in zone i and a rider in zone j, let p_{ij} denote the pre-bonus matching value, which accounts for rider-side revenue, baseline driver compensation, and other non-bonus costs. Thus, the net contribution of this match is $p_{ij} - b_i$, since bonus eligibility is determined by the driver's post-response zone at the time of dispatch rather than the rider's request zone. Although ride-hailing operations are inherently dynamic, the DSMDP can be formulated as a static optimization problem over a single decision period. When forecasts of future system conditions are available, their effects can be incorporated through opportunity-cost adjustments to the matching values p_{ij} (Qin et al. 2020), thereby preserving the static structure of the formulation.

3.2. Mathematical Formulation

To formulate the DSMDP, we characterize the post-response driver supply available for matching under a given driver surge map. Surge bonuses affect driver supply through two behavioral channels. First, a dispatchable driver who observes the posted map may remain in the current zone, relocate to a reachable zone, or become unavailable for dispatch during the current period. For any $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$, let $P_{ij}(\boldsymbol{b})$ denote the probability that a dispatchable driver initially located in zone i is available in zone j after responding to the map. Let $P_i^{\text{Off}}(\boldsymbol{b})$ denote the probability that the driver instead becomes unavailable for dispatch. Note that $i \in \mathcal{Z}_i$, so the choice set includes remaining in the current zone.

We model these choices with an MNL model whose alternatives are the reachable service zones and an unavailable state. Each driver therefore compares the expected attractiveness of remaining dispatchable in a reachable zone with that of leaving the dispatchable pool. We use a location-normalized utility representation so that the coefficient combinations governing these choices can be identified from observed transitions. For a driver initially located in zone i, the unnormalized deterministic utility of choosing destination zone $j \in \mathcal{Z}_i$ is $\beta^{(0)} + \beta^{(1)} b_j + \beta^{(2)} b_i + \beta^{(3)} d_{ij}$, where b_j is the destination-zone bonus, b_i is the origin-zone bonus, and d_{ij} is the travel distance or relocation cost from i to j. The parameter $\beta^{(0)}$ is the common intercept of the dispatchable-zone alternatives, $\beta^{(1)}$ measures sensitivity to the destination-zone bonus, $\beta^{(2)}$ captures the effect of the origin-zone bonus on remaining dispatchable, and $\beta^{(3)}$ measures the disutility of relocation distance. The unnormalized utility of entering the unavailable state is $\alpha_i^{\text{Off}} - \gamma^{\text{Off}} b_i$ for each $i \in \mathcal{Z}$, where α_i^{Off} is the zone-specific baseline attractiveness of becoming unavailable and $\gamma^{\text{Off}} \geq 0$ measures how the origin-zone bonus reduces the attractiveness of this state.

Because MNL choice probabilities are invariant to subtracting the same origin-specific term from all alternatives, subtracting $\beta^{(0)} + \beta^{(2)} b_i$ yields the normalized utilities $V_{ij} = \beta^{(1)} b_j + \beta^{(3)} d_{ij}$

and $V_i^{\mathrm{Off}} = \delta_i^{\mathrm{Off}} - \zeta^{\mathrm{Off}} b_i$, where $\delta_i^{\mathrm{Off}} = \alpha_i^{\mathrm{Off}} - \beta^{(0)}$ denotes the zone-specific relative intercept of the unavailable state and $\zeta^{\mathrm{Off}} = \beta^{(2)} + \gamma^{\mathrm{Off}}$. We impose $\zeta^{\mathrm{Off}} \geq 0$, so that a higher origin-zone bonus weakly reduces the relative attractiveness of becoming unavailable. For simplicity, let $\boldsymbol{\beta} = (\beta^{(1)}, \beta^{(3)})$. This reparameterization preserves all choice probabilities and allows the identified coefficient combinations to be estimated directly. For each $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$, the relocation and unavailability probabilities are $P_{ij}(\boldsymbol{b}) = \exp(V_{ij})/[\sum_{m\in\mathcal{Z}_i} \exp(V_{im}) + \exp(V_i^{\mathrm{Off}})]$ and $P_i^{\mathrm{Off}}(\boldsymbol{b}) = \exp(V_i^{\mathrm{Off}})/[\sum_{m\in\mathcal{Z}_i} \exp(V_{im}) + \exp(V_i^{\mathrm{Off}})]$, respectively, and satisfy $\sum_{j\in\mathcal{Z}_i} P_{ij}(\boldsymbol{b}) + P_i^{\mathrm{Off}}(\boldsymbol{b}) = 1$. For convenience, set $P_{ij}(\boldsymbol{b}) = 0$ whenever $j \notin \mathcal{Z}_i$.

Bonuses may also activate latent drivers, including drivers who may otherwise serve competing platforms. Let $P_i^{\mathrm{On}}(\boldsymbol{b})$ denote the corresponding bonus-dependent activation probability. We model this probability using the binary logit model $P_i^{\mathrm{On}}(\boldsymbol{b}) = \sigma(\alpha_i^{\mathrm{On}} + \gamma^{\mathrm{On}} b_i)$ for each $i \in \mathcal{Z}$, where $\sigma(z) = 1/(1+\exp(-z))$, α_i^{On} is the zone-specific baseline activation propensity, and $\gamma^{\mathrm{On}} \geq 0$ measures the sensitivity of latent-driver activation to the local bonus. In deployment, the identifiable behavioral-response parameters $\boldsymbol{\beta}$, $\boldsymbol{\delta}^{\mathrm{Off}}$, ζ^{Off}, $\boldsymbol{\alpha}^{\mathrm{On}}$, and γ^{On} can be estimated from historical driver-state transitions using the calibration procedure in Section EC.1 of the e-companion. They enter each real-time DSMDP instance as inputs.

Let $\boldsymbol{P}(\boldsymbol{b}) = \{P_{ij}(\boldsymbol{b})\}_{i\in\mathcal{Z},\, j\in\mathcal{Z}_i}$ denote the collection of bonus-induced relocation probabilities. The post-response driver supply in zone i combines the initial dispatchable supply, relocation inflows and outflows, departures from the dispatchable pool, and activation of latent drivers. Specifically, for each $i \in \mathcal{Z}$, the post-response supply is

$$s_i(\boldsymbol{b}) = s_i^{\mathrm{On}} + \sum_{j\in\mathcal{Z}_i\setminus\{i\}} s_j^{\mathrm{On}} P_{ji}(\boldsymbol{b}) - s_i^{\mathrm{On}} \left[\sum_{j\in\mathcal{Z}_i\setminus\{i\}} P_{ij}(\boldsymbol{b}) + P_i^{\mathrm{Off}}(\boldsymbol{b})\right] + s_i^{\mathrm{Off}} P_i^{\mathrm{On}}(\boldsymbol{b}). \tag{1}$$

The five terms in Eq. (1) represent initial dispatchable supply, relocation inflows, relocation outflows, departures from the dispatchable pool, and activation of latent drivers, respectively. Because the response probabilities are nonnegative and sum to one, the initial dispatchable supply in each zone is allocated among drivers who remain, relocate, or become unavailable. The latent-supply term adds the expected number of drivers activated by the local bonus. These two probability allocations imply that the post-response supply is nonnegative. We write $\boldsymbol{s}(\boldsymbol{b}) := (s_i(\boldsymbol{b}))_{i\in\mathcal{Z}}$ for the post-response supply vector.

Based on the post-response supply, the platform assigns drivers to requests. A match from driver zone i to request zone j has pre-bonus value $p_{ij} \geq 0$ and net contribution $p_{ij} - b_i$. The platform limits each zone-level bonus by $b^{\max} > 0$, realized bonus spending by B, and neighboring-zone bonus differences by the spatial smoothness threshold Δ^b; the latter is consistent with ride-hailing

incentive design and observed platform products (Ong et al. 2021). Figure 1 illustrates the spatial incentive maps that motivate this smoothness requirement. The resulting model is

$$\text{(DSMDP)} \quad \max_{\boldsymbol{b},\boldsymbol{x}} \quad \sum_{i,j\in\mathcal{Z}} (p_{ij}-b_i)x_{ij} \tag{2a}$$

$$\text{s.t.} \quad \sum_{i\in\mathcal{Z}} x_{ij} \le d_j, \qquad j\in\mathcal{Z}, \tag{2b}$$

$$\sum_{j\in\mathcal{Z}} x_{ij} \le s_i(\boldsymbol{b}), \qquad i\in\mathcal{Z}, \tag{2c}$$

$$\sum_{i,j\in\mathcal{Z}} b_i x_{ij} \le B, \tag{2d}$$

$$|b_i-b_j| \le \Delta^b, \qquad i\in\mathcal{Z}, \quad j\in\mathcal{Z}_i\setminus\{i\}, \tag{2e}$$

$$x_{ij} \ge 0, \qquad i,j\in\mathcal{Z}, \tag{2f}$$

$$0 \le b_i \le b^{\max}, \qquad i\in\mathcal{Z}. \tag{2g}$$

Objective (2a) maximizes the pre-bonus matching value net of bonus payments. Constraint (2b) limits assignments by rider demand, whereas Constraint (2c) limits them by the supply available after driver response. Constraint (2d) enforces the realized bonus-spending budget, and Constraint (2e) imposes spatial smoothness on the displayed map. Constraints (2f)–(2g) impose nonnegativity on matching flows and bound each zone bonus. Directly solving DSMDP within the short decision window is challenging because bonuses enter the nonlinear driver-response functions and interact bilinearly with matching flows through the payment terms $b_i x_{ij}$ in both the objective and the realized-spending constraint.

4. Marginal Analysis of Zone-Level Bonuses

Solving the DSMDP requires assessing whether changes in zone-level bonuses generate sufficient matching value to justify the associated payments. We therefore characterize the marginal value of a zone bonus, identifying when introducing bonuses is beneficial and how spatial smoothness couples bonus decisions across zones. This characterization also guides the bonus updates in the iterative solution approach developed next.

4.1. Marginal Value of a Zone Bonus

To characterize the marginal value of a zone bonus, we first examine how the downstream matching outcome depends on the driver surge map. For a fixed map $\boldsymbol{b}$, the driver response model determines the available supply $\boldsymbol{s}(\boldsymbol{b})$ through Eq. (1). The DSMDP then reduces to the following LP, which determines the optimal matching outcome conditional on $\boldsymbol{b}$:

$$\text{(DSMDP-M)} \quad Z(\boldsymbol{b}) = \max_{\boldsymbol{x}} \quad \sum_{i,j\in\mathcal{Z}} (p_{ij}-b_i)x_{ij} \tag{3a}$$

$$\text{s.t.} \quad \sum_{i\in\mathcal{Z}} x_{ij} \le d_j, \qquad j\in\mathcal{Z}, \tag{3b}$$

$$\sum_{j\in\mathcal{Z}} x_{ij} \le s_i(\boldsymbol{b}), \qquad i\in\mathcal{Z}, \tag{3c}$$

$$\sum_{i,j\in\mathcal{Z}} b_i x_{ij} \le B \tag{3d}$$

$$x_{ij} \ge 0, \qquad i,j\in\mathcal{Z}. \tag{3e}$$

Here, $Z(\boldsymbol{b})$ denotes the optimal net operating profit under the fixed driver surge map $\boldsymbol{b}$. For a selected optimal primal-dual solution, let $\pi_j \ge 0$, $\eta_i \ge 0$, and $\lambda \ge 0$ be the optimal dual variables for Constraints (3b), (3c), and (3d), respectively, representing the shadow prices of rider demand in zone j, post-response driver supply in zone i, and the bonus budget. The bonus vector $\boldsymbol{b}$ affects the net matching coefficients in Objective (3a), the post-response supply in Constraint (3c), and the bonus-payment coefficients in Constraint (3d). For the selected optimal primal-dual pair, differentiating the Lagrangian with respect to the zone bonus b_m gives

$$g_m := \sum_{i\in\mathcal{Z}} \eta_i \frac{\partial s_i(\boldsymbol{b})}{\partial b_m} - (1+\lambda)\sum_{j\in\mathcal{Z}} x_{mj}, \qquad \forall m\in\mathcal{Z}, \tag{4}$$

where g_m is the marginal value of the zone-m bonus associated with the selected optimal primal-dual solution. We say that the stable optimal-basis condition holds at $\boldsymbol{b}$ if, in a neighborhood of $\boldsymbol{b}$, DSMDP-M has a unique primal-dual solution supported by a primal- and dual-nondegenerate optimal basis, with twice continuously differentiable behavioral-response functions. Under this condition, the envelope formula gives $\boldsymbol{g} = \nabla Z(\boldsymbol{b})$. At points where the optimal basis changes, $Z(\boldsymbol{b})$ may be nondifferentiable, and different optimal primal-dual pairs may yield different vectors $\boldsymbol{g}$. Since η_j is the shadow price of supply in zone j, we therefore define $\bar{\eta}_i := \sum_{j\in\mathcal{Z}_i} P_{ij}(\boldsymbol{b})\eta_j$ as the expected downstream matching value of a dispatchable driver initially in zone i, with zero value assigned to the unavailable state. Expanding the supply derivatives gives $\sum_j \eta_j \frac{\partial s_j}{\partial b_m} = \sum_i s_i^{\mathrm{On}} \sum_{j\in\mathcal{Z}_i} \eta_j \frac{\partial P_{ij}}{\partial b_m} + \eta_m s_m^{\mathrm{Off}} \frac{\partial P_m^{\mathrm{On}}}{\partial b_m}$. Let $\mathbb{1}_{\{\cdot\}}$ denote the indicator function. Holding the selected dual vector $\boldsymbol{\eta}$ fixed, the MNL response derivatives in Section EC.2 of the e-companion imply

$$\sum_{j\in\mathcal{Z}_i} \eta_j \frac{\partial P_{ij}(\boldsymbol{b})}{\partial b_m} = \beta^{(1)} \mathbb{1}_{\{m\in\mathcal{Z}_i\}} P_{im}(\boldsymbol{b})\big(\eta_m - \bar{\eta}_i\big) + \zeta^{\mathrm{Off}} \mathbb{1}_{\{i=m\}} P_i^{\mathrm{Off}}(\boldsymbol{b})\bar{\eta}_i. \tag{5}$$

The first term values attraction to zone m relative to the current choice-set average, whereas the second values retention of an origin-i dispatchable driver when the origin-zone bonus makes unavailability less attractive. Aggregating Eq. (5) across origins and adding latent-driver activation yields the economically interpretable channels below.

For a fixed bonus vector $\boldsymbol{b}$ and a selected optimal primal-dual solution of DSMDP-M, define the matched quantity $q_m := \sum_{j\in\mathcal{Z}} x_{mj}$. The destination-attraction, dispatchable-supply-retention, and latent-driver-activation terms are defined as

$$\begin{aligned}
\mathsf{Dest}_m &:= \beta^{(1)} \sum_{i:\, m\in\mathcal{Z}_i} s_i^{\mathrm{On}} P_{im}(\boldsymbol{b})\big(\eta_m - \bar{\eta}_i\big),\\
\mathsf{Ret}_m &:= \zeta^{\mathrm{Off}} s_m^{\mathrm{On}} P_m^{\mathrm{Off}}(\boldsymbol{b})\bar{\eta}_m,\\
\mathsf{Act}_m &:= \gamma^{\mathrm{On}} s_m^{\mathrm{Off}} P_m^{\mathrm{On}}(\boldsymbol{b})\big(1 - P_m^{\mathrm{On}}(\boldsymbol{b})\big)\eta_m.
\end{aligned} \tag{6}$$

PROPOSITION 1 **(Decomposition of marginal bonus value)**. *For every selected optimal primal-dual solution of DSMDP-M, the marginal bonus value decomposes as*

$$g_m = \mathsf{Dest}_m + \mathsf{Ret}_m + \mathsf{Act}_m - (1+\lambda)q_m, \qquad \forall m \in \mathcal{Z}. \tag{7}$$

Proposition 1 separates the matching value generated by driver responses from the additional payments on existing matches and budget opportunity cost. Destination attraction Dest_m compares the supply shadow value in zone m with the driver's expected downstream matching value across existing choices. For $\beta^{(1)} > 0$, the contribution from origin i to zone m is positive precisely when $\eta_m > \bar{\eta}_i$, provided $s_i^{\mathrm{On}} P_{im}(\boldsymbol{b}) > 0$. Dispatchable-supply retention Ret_m preserves the expected matching value $\bar{\eta}_m$ of drivers who would otherwise become unavailable, whereas latent-driver activation Act_m adds supply directly to zone m, where it is valued at η_m. A higher bonus also increases payments on existing matches by q_m per unit increase, with an additional budget opportunity cost of λq_m. The value of a bonus increase therefore depends on both driver responsiveness and downstream matching opportunities, which local shortage alone does not capture.

If the selected optimal primal-dual solution is supported by a locally valid optimal basis, g_m is the sensitivity associated with that basis. Under the stable optimal-basis condition, this sensitivity equals the derivative $\partial Z(\boldsymbol{b})/\partial b_m$.

4.2. Local Value of Bonus Intervention

Proposition 1 characterizes how a marginal change in a zone bonus affects net operating profit. We next use this decomposition to examine when the platform should begin offering bonuses and how this decision depends on the supply available without incentives and drivers' responsiveness. We use a superscript no for quantities evaluated at the no-bonus map $\mathbf{0}$, with $P_i^{\mathrm{On,no}} := \sigma(\alpha_i^{\mathrm{On}})$. For each zone $i \in \mathcal{Z}$, let $\boldsymbol{e}_i$ have entry 1 at i and 0 elsewhere, so that $b_i\boldsymbol{e}_i$ offers bonus b_i only in zone i. Taking the no-bonus map as the benchmark, the following corollary identifies zones where introducing a small positive bonus increases net operating profit.

COROLLARY 1 **(Local value of the first bonus)**. *Suppose $B > 0$, $b^{\max} > 0$, and $\Delta^b > 0$, and suppose that $Z(\boldsymbol{b})$ is differentiable at $\boldsymbol{b} = \mathbf{0}$. The one-sided marginal value of introducing a bonus is*

$$\frac{\mathrm{d}}{\mathrm{d}b_i} Z(b_i\boldsymbol{e}_i)\bigg|_{b_i=0+} = \mathsf{Dest}_i^{\mathrm{no}} + \mathsf{Ret}_i^{\mathrm{no}} + \gamma^{\mathrm{On}} s_i^{\mathrm{Off}} P_i^{\mathrm{On,no}}\big(1 - P_i^{\mathrm{On,no}}\big)\eta_i^{\mathrm{no}} - q_i^{\mathrm{no}}. \tag{8}$$

If the right-hand side of Eq. (8) is positive for some zone i, then a sufficiently small positive bonus in that zone improves the platform objective, and the no-bonus map is neither locally nor globally optimal. The no-bonus map satisfies the first-order stationarity condition if the right-hand side is nonpositive for every zone and is a strict local maximizer if it is strictly negative for every zone. To make the intervention condition in Corollary 1 explicit in terms of driver composition and activation responsiveness, consider a single-zone setting with no relocation or dispatchable-supply exit. Suppose demand is sufficient to match all available supply and the bonus budget remains slack for bonuses sufficiently close to zero. These assumptions isolate the trade-off between activating additional drivers and paying bonuses on existing matches. Each additional activated driver generates a match with pre-bonus value $p_{ii} > 0$, and the platform's net operating profit is

$$Z_i(b_i) = (p_{ii} - b_i)\left[s_i^{\mathrm{On}} + s_i^{\mathrm{Off}} P_i^{\mathrm{On}}(b_i)\right]. \tag{9}$$

For $s_i^{\mathrm{Off}} > 0$ and $\gamma^{\mathrm{On}} > 0$, differentiating Eq. (9) at zero shows that the initial marginal bonus value is positive if and only if $p_{ii} > p_i^{\mathrm{crit}}$, where

$$p_i^{\mathrm{crit}} := \frac{s_i^{\mathrm{On}}/s_i^{\mathrm{Off}} + P_i^{\mathrm{On,no}}}{\gamma^{\mathrm{On}} P_i^{\mathrm{On,no}}\left(1 - P_i^{\mathrm{On,no}}\right)}. \tag{10}$$

The factor $P_i^{\mathrm{On,no}}(1 - P_i^{\mathrm{On,no}})$ is maximized at $P_i^{\mathrm{On,no}} = 1/2$. Thus, holding γ^{On} fixed, a small bonus has the greatest effect on activation probability when latent drivers are equally likely to participate or remain inactive without incentives.

4.3. Spatial Balance of Marginal Values

Once bonuses are introduced, spatial smoothness limits how the platform can adjust them across zones. If a zone's bonus already exceeds a neighboring zone's bonus by Δ^b, increasing it further requires raising the neighboring bonus as well. We therefore characterize how marginal bonus values balance across zones whose bonus adjustments are linked by spatial smoothness.

Let $\mathcal{E}^b = \left\{\{i,j\} : i \in \mathcal{Z},\ j \in \mathcal{Z}_i \setminus \{i\}\right\}$. Each unordered neighboring-zone pair appears once in $\mathcal{E}^b$. For each edge $e = \{i,j\} \in \mathcal{E}^b$, choose an arbitrary orientation (i,j). Let $\boldsymbol{D}$ be the oriented incidence matrix, with one row per zone pair and one column per zone, and let $\mathbf{1}$ denote an all-ones vector of conformable dimension. For each edge e, using its chosen orientation (i,j), set $D_{ei} = 1$, $D_{ej} = -1$, and all other entries in that row to zero. Thus, $(\boldsymbol{D}\boldsymbol{b})_e = b_i - b_j$. For $0 < \Delta^b < +\infty$, the bonus-feasible set is

$$\mathcal{B} := \left\{\boldsymbol{b} \in \mathbb{R}^{|\mathcal{Z}|} : \mathbf{0} \le \boldsymbol{b} \le b^{\max}\mathbf{1},\ -\Delta^b\mathbf{1} \le \boldsymbol{D}\boldsymbol{b} \le \Delta^b\mathbf{1}\right\}. \tag{11}$$

Consider a locally optimal bonus map $\boldsymbol{b}^\star \in \mathcal{B}$, i.e., there exists a radius $r_{\mathrm{loc}} > 0$ such that $Z(\boldsymbol{b}^\star) \ge Z(\boldsymbol{b})$ for every $\boldsymbol{b} \in \mathcal{B}$ satisfying $\|\boldsymbol{b} - \boldsymbol{b}^\star\|_2 < r_{\mathrm{loc}}$. Assume Z is differentiable at $\boldsymbol{b}^\star$, where $Z(\boldsymbol{b}^\star)$ denotes the optimal value of DSMDP-M under bonus map $\boldsymbol{b}^\star$. Let $\boldsymbol{g}^\star := \nabla Z(\boldsymbol{b}^\star)$, with entries $g_i^\star$ given by Eq. (7). We call zone i interior if $0 < b_i^\star < b^{\max}$. For the constraints defining $\mathcal{B}$, let u_i^+ and

u_i^- denote the nonnegative Karush–Kuhn–Tucker (KKT) multipliers for the upper and lower bonus bounds, respectively. For each edge e, oriented as (i,j), let ρ_e^+ and ρ_e^- denote the nonnegative KKT multipliers for $b_i - b_j \le \Delta^b$ and $b_j - b_i \le \Delta^b$, respectively. These multipliers satisfy complementary slackness with the constraints, which leads to the following balance condition for marginal bonus values.

PROPOSITION 2 **(Spatial balance of marginal bonus values)**. *At $\boldsymbol{b}^\star$, the first-order optimality condition can be written as*

$$g_i^\star = u_i^+ - u_i^- + \sum_{e \in \mathcal{E}^b} D_{ei}(\rho_e^+ - \rho_e^-), \qquad i \in \mathcal{Z}. \tag{12}$$

If zone i is interior and all spatial smoothness multipliers on edges incident to i are zero, then $g_i^\star = 0$. Substituting this into Eq. (7) yields

$$\mathsf{Dest}_i^\star + \mathsf{Ret}_i^\star + \mathsf{Act}_i^\star = (1+\lambda^\star) q_i^\star. \tag{13}$$

In particular, this condition holds when zone i is interior and all of its incident spatial smoothness constraints are slack.

Proposition 2 shows that spatial smoothness couples marginal bonus values across neighboring zones, so an interior zone need not satisfy $g_i^\star = 0$. Define $\mathcal{E}_\rho := \{e \in \mathcal{E}^b : \rho_e^+ + \rho_e^- > 0\}$ and consider the graph $(\mathcal{Z}, \mathcal{E}_\rho)$. The following corollary characterizes how these marginal values balance over each connected component, i.e., a maximal group of zones linked, directly or through other zones, by edges in $\mathcal{E}_\rho$.

COROLLARY 2 **(Balance across connected zones)**. *For any multiplier system satisfying Proposition 2, let C be a connected component of $(\mathcal{Z}, \mathcal{E}_\rho)$. Then*

$$\sum_{i \in C} g_i^\star = \sum_{i \in C} (u_i^+ - u_i^-). \tag{14}$$

If every zone in C is interior, then

$$\sum_{i \in C} [\mathsf{Dest}_i^\star + \mathsf{Ret}_i^\star + \mathsf{Act}_i^\star - (1+\lambda^\star) q_i^\star] = 0. \tag{15}$$

Corollary 2 further shows that spatial smoothness links marginal bonus values across zones, so bonus adjustments must account for the active smoothness constraints that couple neighboring zones. If every zone in a component is interior, positive and negative marginal bonus values offset one another. When $\Delta^b = +\infty$, spatial smoothness constraints are absent, and each interior zone balances independently. This constrained balance motivates ISA, which combines marginal sensitivities with active bonus-map constraints when constructing its update direction.

5. Solution Approach and Analytical Results

The analysis in Section 4 provides marginal bonus values that can guide adjustments to the bonus map. We use this information to develop ISA, an iterative solution approach for the DSMDP. At each iteration, ISA evaluates driver responses and solves DSMDP-M, then uses the resulting sensitivity information to update bonuses subject to the bonus bounds and spatial smoothness constraints. We first present the algorithm, then analyze its stationarity properties and develop an upper bound on the optimal net operating profit.

5.1. Iterative Solution Approach

Let $\boldsymbol{b}^k \in \mathcal{B}$ denote the current bonus map at iteration k, where $\mathcal{B}$ is defined in Eq. (11). As bonuses vary, the optimal matching basis may change, making $Z(\boldsymbol{b})$ nonsmooth. We refer to the objective-value function defined by a fixed optimal matching basis as a matching branch. An adjustment that improves one branch may not improve another, so ISA uses sensitivities from nearby branches to construct its update.

At the current map $\boldsymbol{b}^k$, referred to as the center, ISA samples nearby bonus maps within a sampling radius $\epsilon_k > 0$. These ambient samples are drawn from the surrounding bonus space without being restricted to $\mathcal{B}$. Let $\mathcal{A}_k$ index the ambient samples, and let $\boldsymbol{y}^{k,a}$, $a \in \mathcal{A}_k$, denote the corresponding sampled bonus maps. Set $\boldsymbol{y}^{k,0} := \boldsymbol{b}^k$ and $\mathcal{I}_k := \{0\} \cup \mathcal{A}_k$. At each $\boldsymbol{y}^{k,a}$, $a \in \mathcal{I}_k$, ISA evaluates driver responses, solves DSMDP-M, and uses the optimal primal-dual solution to compute the corresponding branch sensitivity $\boldsymbol{g}^{k,a}$ from Eq. (7). For the center sensitivity, write $\boldsymbol{g}^k := \boldsymbol{g}^{k,0}$. The matrix $\boldsymbol{G}_k = [\boldsymbol{g}^{k,a}]_{a \in \mathcal{I}_k}$ collects these sensitivities and forms the working bundle. For later stationarity assessment, we use the ambient sensitivity matrix $\boldsymbol{G}_k^{\text{amb}} := [\boldsymbol{g}^{k,a}]_{a \in \mathcal{A}_k}$, which excludes the center evaluation.

To construct a direction that respects the bonus constraints, let $\boldsymbol{N}_k := [\boldsymbol{n}^{k,c}]_{c \in \mathcal{C}_k}$ collect the outward normals of all constraints defining $\mathcal{B}$ that are active at $\boldsymbol{b}^k$, where $\mathcal{C}_k$ indexes these constraints and $\boldsymbol{n}^{k,c}$ is the outward normal of constraint c. Let $\boldsymbol{\omega}$ denote the convex-combination weights on sampled sensitivities, with $\boldsymbol{\omega} \geq 0$ and $\mathbf{1}^\top \boldsymbol{\omega} = 1$, and let $\boldsymbol{\mu} \geq 0$ denote the weights on the columns of $\boldsymbol{N}_k$. ISA selects a convex combination $\boldsymbol{G}_k \boldsymbol{\omega}$ of sampled sensitivities and a nonnegative combination $\boldsymbol{N}_k \boldsymbol{\mu}$ of active constraint normals by solving the quadratic program (QP)

$$(\boldsymbol{\omega}^k, \boldsymbol{\mu}^k) \in \arg \min_{\substack{\boldsymbol{\omega} \geq \mathbf{0},\ \mathbf{1}^\top \boldsymbol{\omega} = 1 \\ \boldsymbol{\mu} \geq \mathbf{0}}} \frac{1}{2} \left\| \boldsymbol{G}_k \boldsymbol{\omega} - \boldsymbol{N}_k \boldsymbol{\mu} \right\|_2^2 . \tag{16}$$

The QP finds the closest pair of these combinations. Define $\boldsymbol{r}^k := \boldsymbol{G}_k \boldsymbol{\omega}^k - \boldsymbol{N}_k \boldsymbol{\mu}^k$ as the resulting constraint-adjusted direction at iteration k and $\widehat{R}_k := \|\boldsymbol{r}^k\|_2$ as the working-bundle residual. When $\boldsymbol{r}^k \neq \mathbf{0}$, the QP optimality conditions give $\boldsymbol{N}_k^\top \boldsymbol{r}^k \leq \mathbf{0}$ and $(\boldsymbol{g}^{k,a})^\top \boldsymbol{r}^k \geq \|\boldsymbol{r}^k\|_2^2 > 0$ for every $a \in \mathcal{I}_k$. Thus, the direction respects the active constraints and predicts a positive improvement for every

sampled branch. The step size is then determined by evaluating the actual change in net operating profit.

For a candidate step size $\tau > 0$, ISA projects $\boldsymbol{b}^k + \tau \boldsymbol{r}^k$ onto $\mathcal{B}$. Let $\Pi_{\mathcal{B}}$ denote this Euclidean projection and define the feasible trial displacement as $\boldsymbol{h}_k(\tau) := \Pi_{\mathcal{B}}(\boldsymbol{b}^k + \tau \boldsymbol{r}^k) - \boldsymbol{b}^k$. The trial bonus map is therefore $\boldsymbol{b}^k + \boldsymbol{h}_k(\tau)$. The projection enforces the bonus bounds and spatial smoothness constraints. ISA evaluates driver responses under the trial map and solves DSMDP-M to obtain its net operating profit while enforcing the realized bonus-spending budget.

To obtain sufficient improvement while avoiding a time-consuming exact line search, we select the step size using Armijo backtracking. Let $\chi \in (0,1)$ denote the Armijo sufficient-increase parameter. The projected bundle Armijo test accepts the trial when $\min_{a \in \mathcal{I}_k} (\boldsymbol{g}^{k,a})^\top \boldsymbol{h}_k(\tau) > 0$ and

$$Z\big(\boldsymbol{b}^k + \boldsymbol{h}_k(\tau)\big) \geq Z(\boldsymbol{b}^k) + \chi \min_{a \in \mathcal{I}_k} (\boldsymbol{g}^{k,a})^\top \boldsymbol{h}_k(\tau). \tag{17}$$

Thus, all sampled sensitivities must predict an improvement, and the realized improvement must be at least a fraction χ of the smallest predicted improvement. ISA reduces the step size when a trial fails this test. Otherwise, the trial becomes the new center. If no trial is accepted within the line-search limits, ISA retains the current center and adds new samples, producing a null step.

To determine whether to terminate, ISA first checks the working-bundle residual $\widehat{R}_k$ against a stationarity scale $\nu_k > 0$. A small $\widehat{R}_k$ indicates that the convex combination $\boldsymbol{G}_k \boldsymbol{\omega}^k$ of sampled sensitivities nearly matches the nonnegative combination $\boldsymbol{N}_k \boldsymbol{\mu}^k$ of active constraint normals, i.e., $\|\boldsymbol{G}_k \boldsymbol{\omega}^k - \boldsymbol{N}_k \boldsymbol{\mu}^k\|_2 \approx 0$. However, this balance may be affected by the center sensitivity $\boldsymbol{g}^{k,0}$, as Z can be nonsmooth at $\boldsymbol{b}^k$. ISA therefore performs an ambient-only stationarity assessment by replacing $\boldsymbol{G}_k$ with $\boldsymbol{G}_k^{\mathrm{amb}}$ in Eq. (16). The resulting ambient residual $\widehat{R}_k^{\mathrm{amb}}$, defined in Eq. (19), is used to determine whether to refine the sampling and stationarity scales or terminate. Complete sampling, line-search, and stopping procedures are provided in Section EC.3 of the e-companion. Algorithm 1 summarizes ISA, which returns the best feasible incumbent.

5.2. Theoretical Analysis

We next examine the theoretical properties of the proposed solution approach. We first establish the existence of an optimal solution and show how the sampling radius and ambient residual provide a finite-scale stationarity certificate. We then study stationarity along ambient assessments with vanishing sampling radii and residuals. Complete proofs are provided in Section EC.4 of the e-companion.

5.2.1. Existence and Stationarity. For each fixed bonus vector, DSMDP-M is solved to optimality, and $Z(\boldsymbol{b})$ denotes its optimal value. The optimization problem and Algorithm 1 use a positive budget B and solve $\max_{\boldsymbol{b} \in \mathcal{B}} Z(\boldsymbol{b})$. The set $\mathcal{B}$ is nonempty, compact, and convex because it is defined by bonus bounds and linear spatial smoothness constraints.

Algorithm 1 Iterative solution approach for DSMDP.

Input: Initial map $\boldsymbol{b}^0 \in \mathcal{B}$, sampling scales, line-search parameters, and computation limits

Output: Final bonuses $\boldsymbol{b}^{\text{best}}$ and matching decision $\boldsymbol{x}^{\text{best}}$

Initialize $k \leftarrow 0$, the sample pool, and $Z^{\text{best}} \leftarrow -\infty$
while the computation limits remain **do**
  (Center evaluation) Compute driver response and supply at $\boldsymbol{b}^k$, and solve DSMDP-M in Eqs. (3a)–(3e) for $Z(\boldsymbol{b}^k)$, $\boldsymbol{x}^k$, the dual multipliers, and $\boldsymbol{g}^k$
  if $Z(\boldsymbol{b}^k) > Z^{\text{best}}$ **then** set $(Z^{\text{best}}, \boldsymbol{b}^{\text{best}}, \boldsymbol{x}^{\text{best}}) \leftarrow (Z(\boldsymbol{b}^k), \boldsymbol{b}^k, \boldsymbol{x}^k)$
  (Sensitivity sampling) Evaluate nearby matching branches and form $\boldsymbol{G}_k$ and $\boldsymbol{N}_k$
  (Constrained direction) Solve Eq. (16) for $\boldsymbol{r}^k$ and $\widehat{R}_k$
  Perform the ambient-only assessment when required (see Section EC.3.3 of the e-companion).
  (Line search) Apply Eq. (17) with $\boldsymbol{h}_k(\tau) = \Pi_{\mathcal{B}}(\boldsymbol{b}^k + \tau \boldsymbol{r}^k) - \boldsymbol{b}^k$
  if the trial is accepted **then**
    Set $\boldsymbol{b}^{k+1} \leftarrow \boldsymbol{b}^k + \boldsymbol{h}_k(\tau)$, update the best incumbent, and set $k \leftarrow k+1$
    Rebuild the sample pool around the new center
  else
    Retain the center and k, and enrich the sample pool
return the best feasible incumbent $(\boldsymbol{b}^{\text{best}}, \boldsymbol{x}^{\text{best}})$

LEMMA 1 **(Continuity and existence).** *Suppose that $P_{ij}(\boldsymbol{b})$, $P_i^{\text{On}}(\boldsymbol{b})$, and $P_i^{\text{Off}}(\boldsymbol{b})$ are continuous on an open set containing $\mathcal{B}$. Then $Z(\boldsymbol{b})$ is continuous on $\mathcal{B}$, and the reduced DSMDP $\max_{\boldsymbol{b}\in\mathcal{B}} Z(\boldsymbol{b})$ admits a globally optimal solution.*

Lemma 1 shows that small changes in bonuses lead to small changes in optimal net operating profit, even when the optimal matching decisions change. We next examine how ISA can assess stationarity using the sensitivities obtained from nearby bonus maps.

For the finite-scale analysis, suppose that Z admits a locally Lipschitz extension around $\mathcal{B}$. For $\epsilon > 0$, define the enlarged Clarke generalized gradient

$$\partial_\epsilon^C Z(\boldsymbol{b}) := \operatorname{cl}\operatorname{conv}\left(\bigcup_{\|\boldsymbol{y}-\boldsymbol{b}\|_2 \le \epsilon} \partial^C Z(\boldsymbol{y}) \right), \tag{18}$$

where cl conv denotes the closed convex hull, the union is taken over points in the domain of the extension, and $\partial^C Z(\boldsymbol{y})$ denotes its Clarke generalized gradient. For two nonempty sets A_1 and A_2, let $\operatorname{dist}(A_1, A_2) := \inf\{\|\boldsymbol{a}_1 - \boldsymbol{a}_2\|_2 : \boldsymbol{a}_1 \in A_1, \boldsymbol{a}_2 \in A_2\}$. Let $\mathcal{N}_{\mathcal{B}}(\boldsymbol{b})$ denote the outward normal cone $\{\boldsymbol{n} \in \mathbb{R}^{|\mathcal{Z}|} : \boldsymbol{n}^\top(\widetilde{\boldsymbol{b}} - \boldsymbol{b}) \le 0,\ \forall \widetilde{\boldsymbol{b}} \in \mathcal{B}\}$. In particular, at the current map $\boldsymbol{b}^k$, $\mathcal{N}_{\mathcal{B}}(\boldsymbol{b}^k) = \{\boldsymbol{N}_k \boldsymbol{\mu} : \boldsymbol{\mu} \ge \boldsymbol{0}\}$. For $\epsilon > 0$ and $\nu \ge 0$, a feasible map $\boldsymbol{b}$ is (ϵ, ν)-stationary if $\operatorname{dist}(\partial_\epsilon^C Z(\boldsymbol{b}), \mathcal{N}_{\mathcal{B}}(\boldsymbol{b})) \le \nu$. We next show that a sufficiently small ambient residual certifies approximate stationarity of the bonus map at the given sampling radius.

PROPOSITION 3 **(Finite-scale stationarity certificate).** *At a center $\boldsymbol{b}^k \in \mathcal{B}$, let $\mathcal{A}_k \neq \varnothing$, and suppose the ambient samples $\boldsymbol{y}^{k,a}$ are differentiability points of the extension satisfying $\|\boldsymbol{y}^{k,a} -$*

$\boldsymbol{b}^k\|_2 \le \epsilon_k$ *and* $\boldsymbol{g}^{k,a} = \nabla Z(\boldsymbol{y}^{k,a})$, $\forall a \in \mathcal{A}_k$. *Let* $\boldsymbol{G}_k^{\mathrm{amb}}$ *be the ambient sensitivity matrix, and let the columns of* $\boldsymbol{N}_k$ *be exactly the active outward normals of* $\mathcal{B}$ *at* $\boldsymbol{b}^k$*, up to duplicate generators. Define the ambient residual as the QP residual obtained from the ambient sensitivity matrix*

$$\widehat{R}_k^{\mathrm{amb}} := \min_{\substack{\boldsymbol{\omega}\ge\mathbf{0},\ \mathbf{1}^\top\boldsymbol{\omega}=1 \\ \boldsymbol{\mu}\ge\mathbf{0}}} \left\| \boldsymbol{G}_k^{\mathrm{amb}}\boldsymbol{\omega} - \boldsymbol{N}_k\boldsymbol{\mu} \right\|_2 . \tag{19}$$

Then

$$\mathrm{dist}\left(\partial^C_{\epsilon_k} Z(\boldsymbol{b}^k), \mathcal{N}_{\mathcal{B}}(\boldsymbol{b}^k)\right) \le \widehat{R}_k^{\mathrm{amb}}. \tag{20}$$

Consequently, for any $\nu_k \ge 0$, $\widehat{R}_k^{\mathrm{amb}} \le \nu_k$ *certifies* (ϵ_k, ν_k)*-stationarity of the current map.*

Proposition 3 shows that the ambient residual $\widehat{R}_k^{\mathrm{amb}}$ provides a certificate of approximate stationarity. The sampling radius ϵ_k determines how far the sampled bonus maps may lie from the current map $\boldsymbol{b}^k$, while the stationarity scale ν_k sets the acceptable upper bound on the ambient residual $\widehat{R}_k^{\mathrm{amb}}$. The result applies to the ambient-only assessment because every sampled sensitivity is a gradient at a nearby differentiability point and $\boldsymbol{N}_k$ contains all active-constraint normals. Since the working bundle also includes the center sensitivity, which may not be a gradient at a nonsmooth matching transition, a small working-bundle residual triggers this separate assessment. We next examine the limiting behavior of the ambient-only assessments as the sampling radius and residual vanish.

Proposition 4 **(Clarke stationarity of sampled limits)**. *Consider a sequence of ambient-only assessments satisfying the conditions of Proposition 3. If* $\boldsymbol{b}^\ell \to \bar{\boldsymbol{b}}, \epsilon_\ell \downarrow 0, \widehat{R}_\ell^{\mathrm{amb}} \to 0$*, then*

$$0 \in -\partial^C Z(\bar{\boldsymbol{b}}) + \mathcal{N}_{\mathcal{B}}(\bar{\boldsymbol{b}}).$$

Hence, $\bar{\boldsymbol{b}}$ *is a constrained Clarke-stationary point of DSMDP.*

Proposition 4 shows that a convergent sequence of bonus maps from ambient-only assessments has a constrained Clarke-stationary limit when the sampling radius and ambient residual vanish. A smaller sampling radius restricts the assessment to bonus maps closer to the current map, while a smaller ambient residual provides a tighter stationarity certificate at that sampling radius. Together with Proposition 3, this result supports ISA's scale-refinement and stopping rules: the ambient residual test $\widehat{R}_k^{\mathrm{amb}} \le \nu_k$ certifies (ϵ_k, ν_k)-stationarity, and scale refinement reduces the sampling radius ϵ_k and stationarity scale ν_k. Termination upon passing this test at the target sampling radius $\epsilon_{\mathrm{tar}} > 0$ and stationarity scale $\nu_{\mathrm{tar}} > 0$ provides an $(\epsilon_{\mathrm{tar}}, \nu_{\mathrm{tar}})$-stationarity certificate for the returned bonus map.

5.3. Conic Outer Relaxation and Upper Bound

To assess the quality of the feasible solutions produced by ISA, we construct a conic outer relaxation $\overline{\text{DSMDP}}$ that provides an upper bound on the optimal net operating profit. The relaxation uses exponential-cone constraints and McCormick envelopes to relax the driver-response relations and bonus-payment products.

For the MNL model, introduce variables t_{ij} and t_i^{Off} to represent $\exp(V_{ij})$ and $\exp(V_i^{\text{Off}})$, respectively, for $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$. Define the denominator variable $D_i = \sum_{m \in \mathcal{Z}_i} t_{im} + t_i^{\text{Off}}$. Let $\underline{V}_{ij}$ and $\overline{V}_{ij}$ denote valid lower and upper bounds on V_{ij} over $\mathcal{B}$, and let $\mathcal{S}_{\exp}(v; \ell, u)$ denote the affine secant of the exponential function over $[\ell, u]$, with $\mathcal{S}_{\exp}(v; \ell, \ell) = \exp(\ell)$. For each $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$, we relax $t_{ij} = \exp(V_{ij})$ using

$$\exp(V_{ij}) \le t_{ij} \le \mathcal{S}_{\exp}\big(V_{ij}; \underline{V}_{ij}, \overline{V}_{ij}\big). \tag{21}$$

The lower inequality is representable using the exponential cone, while the upper inequality follows from convexity of the exponential function. The same construction applies to t_i^{Off}.

We treat P_{ij} and P_i^{Off} as decision variables and relax the bilinear equalities $t_{ij} = P_{ij} D_i$ and $t_i^{\text{Off}} = P_i^{\text{Off}} D_i$ using McCormick envelopes over valid variable bounds. The probability normalization $\sum_{j \in \mathcal{Z}_i} P_{ij} + P_i^{\text{Off}} = 1$ and nonnegativity constraints are retained. For latent-driver activation, use the same procedure to bound the exponential term in the binary logit denominator and use a McCormick envelope to relax the equality between the probability–denominator product and one.

For bonus payments, recall the matched quantity $q_i = \sum_{j \in \mathcal{Z}} x_{ij}$, and introduce w_i to represent $b_i q_i$. Let $\bar{q}_i$ be a valid upper bound on q_i, obtained from total demand and the maximum possible post-response supply in zone i. Over $(b_i, q_i) \in [0, b^{\max}] \times [0, \bar{q}_i]$, the McCormick envelope is

$$\begin{aligned} &w_i \ge 0, \qquad && w_i \ge b^{\max} q_i + \bar{q}_i b_i - b^{\max} \bar{q}_i, \\ &w_i \le b^{\max} q_i, && w_i \le \bar{q}_i b_i, \end{aligned} \qquad i \in \mathcal{Z}. \tag{22}$$

The relaxed objective is $\sum_{i,j \in \mathcal{Z}} p_{ij} x_{ij} - \sum_{i \in \mathcal{Z}} w_i$, and the budget constraint becomes $\sum_{i \in \mathcal{Z}} w_i \le B$. The post-response supply constraints are linear in the probability variables, while the bonus bounds and spatial smoothness constraints remain $\boldsymbol{b} \in \mathcal{B}$. The variable bounds and complete conic formulation are provided in Sections EC.5.1–EC.5.5 of the e-companion.

Let $Z^\star$ and $\overline{Z}$ denote the optimal objective values of DSMDP and $\overline{\text{DSMDP}}$, respectively. Every feasible DSMDP solution satisfies the exponential bounds and McCormick envelopes when the auxiliary variables equal the quantities they represent. It therefore has a feasible representation in the conic relaxation with the same objective value, giving $Z^\star \le \overline{Z}$, as proved in Section EC.5.6 of the e-companion. If the relaxation is not solved to optimality, the solver-certified upper bound U^{solver} remains valid, so that $Z^\star \le \overline{Z} \le U^{\text{solver}}$. The difference between U^{solver} and the net operating profit of ISA's best feasible incumbent therefore bounds its optimality gap. To obtain a feasible Conic benchmark, we project the relaxation incumbent's bonus map onto $\mathcal{B}$, recompute driver responses, and solve DSMDP-M, as described in Section EC.5.7 of the e-companion.

6. Computational Study

We use generated instances and a New York City case study to address five questions. First, we examine whether offering zone-level bonuses improves the platform's net operating profit by comparing the resulting solutions with the no-bonus benchmark $\mathbf{B}^0$. Second, we investigate how the bonus budget affects realized spending and net operating profit, thereby assessing the value of allocating additional funds to driver incentives. Third, we examine how driver responsiveness, behavioral-parameter estimation error, and spatial smoothness affect the benefits of bonus intervention. Fourth, we evaluate whether ISA achieves higher net operating profit than the feasible benchmark methods on instances of operational scale under the specified timing protocol. Finally, we use the New York City case study to examine ISA's large-scale performance and the spatial allocation of bonuses, using data-derived geography, demand, and a proxy for dispatchable supply. These analyses also examine how the marginal bonus values and spatial balance characterized in Section 4 help explain where bonuses are allocated and when they improve net operating profit.

6.1. Instance Generation

To evaluate different system configurations under controlled conditions, we conduct experiments on generated ride-hailing instances. Each instance represents a single real-time decision snapshot of a spatial ride-hailing market. Following spatial ride-hailing models (Bimpikis et al. 2019), we construct each market on a rows $\times$ columns grid, where each cell represents a service zone and d_{ij} is the Euclidean distance between zone centers. Rider demand is generated from a base level and nonoverlapping spatial hotspots following Soza-Parra et al. (2024), with hotspot intensity and dispersion controlled by θ^d and σ^d. Initial dispatchable driver supply is distributed around activity centers and locally shifted toward demand-concentrated areas, producing spatial supply–demand mismatches (Ong et al. 2021). We control the aggregate market condition through $\kappa = \sum_{i\in\mathcal{Z}} s_i^{\mathrm{On}} / \sum_{j\in\mathcal{Z}} d_j$.

Latent supply is generated to be correlated with demand, while pre-bonus matching values combine pickup-distance decay and destination attractiveness. For each zone i, $\mathcal{Z}_i$ denotes the 3×3 neighborhood used to define short-term relocation options and spatial smoothness constraints. Detailed construction procedures and parameter settings are provided in Section EC.6.1 of the e-companion.

6.2. Benchmark Methods

We compare ISA with five feasible benchmarks motivated by ride-hailing incentives, matching optimization, and nonlinear optimization (Lu et al. 2018, Bimpikis et al. 2019, Besbes et al. 2021, Xu et al. 2018, Özkan and Ward 2020). 1) $\mathbf{B}^0$: The platform posts no bonuses. 2) $\mathbf{H}_1$: This heuristic constructs a one-pass bonus map from local shortage, matching value, and MNL responsiveness.

3) $\mathbf{H}_2$: This heuristic constructs a one-pass bonus map from the dual sensitivities of the no-bonus matching LP. 4) NOMAD: We apply derivative-free local search to the bonus vector (Le Digabel 2011). 5) Conic: We project the bonus vector obtained from the conic outer relaxation in Section 5.3 onto $\mathcal{B}$, recompute driver responses, and solve DSMDP-M to obtain a feasible solution. We also report UB, the solver-certified upper bound for the conic outer relaxation. In the result tables, Obj. denotes the objective value and Time reports wall-clock seconds. The green percentages beside ISA's objective values report its improvement over the strongest non-ISA feasible benchmark. All method entries are feasible; UB is excluded from method comparisons. Detailed benchmark procedures are provided in Section EC.6 of the e-companion.

All experiments are run on a workstation equipped with an AMD Ryzen 9 9950X3D processor (16 physical cores and 32 hardware threads) and 64 GB of RAM. To reflect practical time constraints, we set a 120-second optimization time limit for $\mathbf{H}_1$, $\mathbf{H}_2$, ISA, and NOMAD. The reported runtimes include all computational steps and may therefore slightly exceed this limit. Table 9 reports cumulative runtime across the initial 120-second phase and the extended restart.

6.3. Value of Intervention

We fix $|\mathcal{Z}| = 150$ and vary the initial dispatchable-supply-to-demand ratio κ from 0.20 to 1.20, covering settings from severe shortage to oversupply while holding all other parameters fixed. Table 1 supports two distinct comparisons. Relative to the no-bonus policy $\mathbf{B}^0$, the driver surge map obtained by ISA improves the objective by 62.17% at $\kappa = 0.20$; this intervention value decreases steadily to 12.99% at $\kappa = 1.20$. Within these modeled settings, bonuses therefore generate the greatest incremental value when immediately dispatchable supply is scarce. This pattern is consistent with Corollary 1: when more drivers are already matched in a zone at the no-bonus map, a small bonus increase entails larger payments on existing matches. Introducing a bonus is therefore worthwhile only when the matching value generated through destination attraction, retention, and activation exceeds these additional payments.

The percentages reported alongside the ISA objectives instead compare the complete feasible methods, excluding UB. ISA attains the best objective in every row and improves upon the strongest non-ISA feasible benchmark by 4.82%–16.46%, with the largest method-level differences occurring under moderate shortage. ISA requires 29.04–33.72 seconds, whereas NOMAD nearly exhausts its 120-second limit.

6.4. Value of the Bonus Budget

To quantify the operational value of driver surge maps and examine how this value changes with the available bonus budget, we vary the budget over $B \in \{0, 300, 500, 1000, 1200\}$ for instances with $|\mathcal{Z}| = 150$, holding all other parameters fixed. For each setting, we report the objective value (Obj.),

Table 1 Performance under different supply–demand regimes.

Regime	κ	Conic		$\mathbf{B}^0$		$\mathbf{H}_1$		$\mathbf{H}_2$		NOMAD		UB		ISA	
		Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time
Severe shortage	0.20	4432.36	1.55	5815.88	0.08	6453.35	0.17	6234.41	0.24	8997.40	119.48	14318.88	1.47	**9431.46** ↑4.82%	33.72
	0.30	4139.56	1.51	7736.52	0.07	7549.75	0.18	8111.77	0.24	10428.60	119.76	16335.96	1.45	**11281.64** ↑8.18%	29.37
Moderate shortage	0.40	3800.15	1.51	9353.77	0.08	8244.08	0.18	9698.14	0.25	11172.96	119.76	17982.31	1.45	**12832.67** ↑14.85%	32.84
	0.50	3469.64	1.41	10804.30	0.07	9037.60	0.18	11111.25	0.25	12179.25	119.76	19403.40	1.33	**14183.46** ↑16.46%	30.84
Mild shortage	0.60	3150.65	1.48	12101.25	0.07	10106.02	0.19	12370.68	0.23	14037.15	119.77	20554.12	1.42	**15408.77** ↑9.77%	32.30
	0.70	2861.47	1.62	13274.30	0.09	11075.36	0.19	13498.27	0.26	15147.18	119.77	21333.38	1.55	**16459.56** ↑8.66%	32.92
Near balanced	0.80	2622.43	1.41	14310.55	0.09	11943.39	0.19	14487.80	0.26	16059.66	119.77	21647.29	1.32	**17368.96** ↑8.15%	29.87
	0.90	2427.80	1.46	15220.59	0.09	12524.17	0.17	15351.59	0.25	17059.60	119.77	21875.30	1.38	**18125.72** ↑6.25%	32.26
	1.00	2286.11	1.41	16003.50	0.09	13145.11	0.19	16003.50	0.16	17699.53	119.77	22076.40	1.35	**18630.80** ↑5.26%	29.04
Oversupplied	1.10	2143.48	1.39	16567.67	0.09	13813.78	0.17	16567.67	0.18	17951.07	119.65	22250.86	1.30	**18842.11** ↑4.96%	29.97
	1.20	2024.37	1.26	16852.29	0.09	14082.46	0.19	16852.29	0.17	18148.77	119.77	22400.38	1.18	**19040.79** ↑4.92%	30.54

objective improvement (Obj. Imp.), defined as $(\mathrm{Obj}(B) - \mathrm{Obj}(0))/\mathrm{Obj}(0) \times 100\%$, and realized bonus spending (Spend.). We also report the match rate, defined as $\sum_{i\in\mathcal{Z}}\sum_{j\in\mathcal{Z}} x_{ij} / \sum_{j\in\mathcal{Z}} d_j \times 100\%$, and return on investment (ROI), defined as $(\mathrm{Obj}(B) - \mathrm{Obj}(0))/\mathrm{Spend}(B) \times 100$. Table 2 shows that larger bonus budgets improve the attained objective and match rate in the tested settings, while ROI declines as spending increases. At $B = 1200$, the objective improvement is 42.17% under $\kappa = 0.50$, compared with 16.63% under $\kappa = 1.10$, indicating that incentives are most valuable when supply is scarce. Moreover, realized bonus spending is below the nominal budget of $B = 1200$ in the balanced and oversupplied settings. These results suggest that the platform can use the incremental net operating profit from budget expansion to select an appropriate bonus budget, identifying when further increases in spending deliver only modest additional gains.

Table 2 Value of driver surge maps under different bonus budgets.

κ	Bonus budget	Obj.	Obj. Imp. (%)	Spend.	Match rate (%)	ROI (%)
0.50	0	10804	0.00	0.00	45.59	–
	300	13367	23.72	300.00	54.51	854.27
	500	14183	31.28	500.00	57.74	675.83
	1000	15190	40.60	1000.00	62.72	438.61
	1200	15360	42.17	1199.99	64.04	379.64
1.00	0	16004	0.00	0.00	87.61	–
	300	18067	12.90	300.00	96.30	687.99
	500	18631	16.42	500.00	99.30	525.46
	1000	19106	19.39	1000.00	100.00	310.26
	1200	19144	19.63	1189.80	100.00	263.98
1.10	0	16568	0.00	0.00	96.01	–
	300	18352	10.77	300.00	100.00	594.75
	500	18842	13.73	500.00	100.00	454.89
	1000	19296	16.47	1000.00	100.00	272.81
	1200	19323	16.63	1155.99	100.00	238.34

6.5. Behavioral and Spatial Sensitivity Analyses

We next examine how latent-driver activation responsiveness, behavioral-parameter estimation error, and spatial smoothness affect the benefits of bonus intervention. Each analysis varies one factor while holding the remaining experimental settings fixed.

6.5.1. Responsiveness of Latent Driver Activation. We vary γ^{On} over $\{1.0, 1.2, 1.5, 1.8, 2.0\}$ while holding all other parameters fixed; $\gamma^{\mathrm{On}} = 1.5$ is the baseline, and higher values represent stronger activation responses among latent drivers. Table 3 shows that stronger activation responsiveness is associated with higher attained objective and ROI across the tested settings. As γ^{On} increases from 1.0 to 2.0, the objective improvement rises from 23.07% to 37.54% with $\kappa = 0.50$ and from 9.95% to 16.55% with $\kappa = 1.10$. The objective can continue to increase even when the aggregate match rate has reached 100%, because newly dispatchable supply may support higher-value matches. Equation (10) shows that, holding the baseline activation probability fixed, stronger activation responsiveness lowers the intervention threshold. A small initial bonus improves net operating profit when the pre-bonus matching value exceeds this threshold. The value of activating latent drivers therefore depends on both their responsiveness and the matching opportunities available to the platform.

Table 3 Sensitivity analysis with respect to latent-driver activation responsiveness.

κ	γ^{On}	Obj.	Obj. Imp. (%)	Spend.	Match rate (%)	ROI (%)
0.50	1.0	13297	23.07	500.00	54.91	498.46
	1.2	13683	26.64	500.00	56.13	575.76
	1.5	14183	31.28	500.00	57.74	675.83
	1.8	14609	35.21	500.00	59.09	760.88
	2.0	14860	37.54	500.00	59.91	811.09
1.00	1.0	17936	12.08	500.00	96.50	386.49
	1.2	18251	14.05	500.00	97.76	449.59
	1.5	18631	16.42	500.00	99.30	525.46
	1.8	18931	18.29	500.00	100.00	585.50
	2.0	19103	19.37	500.00	100.00	619.99
1.10	1.0	18216	9.95	500.00	100.00	329.71
	1.2	18502	11.67	500.00	100.00	386.81
	1.5	18842	13.73	500.00	100.00	454.89
	1.8	19144	15.55	500.00	100.00	515.25
	2.0	19309	16.55	500.00	100.00	548.25

6.5.2. Error in Behavioral Parameter Estimates. In deployment, behavioral-response parameters calibrated from historical data may differ from current driver response because of limited observations and changing market conditions. To examine the impact of these inaccuracies, we use the instances with $\kappa \in \{0.50, 1.00, 1.10\}$ from Section 6.3 and independently perturb the six common behavioral coefficients $\beta^{(1)}, \beta^{(3)}, \delta^{\mathrm{Off}}, \zeta^{\mathrm{Off}}, \alpha^{\mathrm{On}}$, and γ^{On} by factors drawn from Uniform$(1-r, 1+r)$, where $r \in \{10\%, 20\%, 30\%, 40\%, 50\%\}$. For these instances, $\delta_i^{\mathrm{Off}} \equiv \delta^{\mathrm{Off}}$ and $\alpha_i^{\mathrm{On}} \equiv \alpha^{\mathrm{On}}$ across all zones. For each (κ, r), ISA constructs driver surge maps using 10 independently perturbed parameter vectors, which are then evaluated under the nominal response. Let Z_κ^{nom} denote the nominal-parameter objective and $Z_{r,\kappa,n}^{\mathrm{deploy}}$ the deployed objective under perturbation n; we define $\mathrm{Loss}_{r,\kappa,n} =$

$(Z_\kappa^{\text{nom}} - Z_{r,\kappa,n}^{\text{deploy}})/|Z_\kappa^{\text{nom}}| \times 100\%$. Table 4 reports the mean deployed objective and the mean, standard deviation, and maximum loss across the 10 perturbations. The mean loss remains between 0.24% and 3.25% across all settings. The largest mean losses occur under supply shortage and at the higher perturbation levels, indicating that accurate behavioral calibration is most valuable when supply is scarce.

Table 4 Sensitivity analysis with respect to behavioral-parameter estimation error.

κ	Perturbation level	Mean Obj.	Mean loss (%)	Std. loss (%)	Max loss (%)
0.50	0%	14183	0.00	–	–
	±10%	14118	0.46	0.59	2.08
	±20%	14058	0.89	0.71	2.12
	±30%	13833	2.47	1.70	5.20
	±40%	13840	2.42	2.04	7.63
	±50%	13722	3.25	2.49	7.64
1.00	0%	18631	0.00	–	–
	±10%	18586	0.24	0.16	0.49
	±20%	18540	0.49	0.28	0.99
	±30%	18502	0.69	0.50	1.55
	±40%	18313	1.70	1.09	3.80
	±50%	18223	2.19	1.14	4.45
1.10	0%	18842	0.00	–	–
	±10%	18788	0.29	0.31	0.96
	±20%	18717	0.66	0.48	1.33
	±30%	18740	0.54	0.50	1.81
	±40%	18594	1.32	1.02	3.26
	±50%	18676	0.88	0.52	2.13

6.5.3. Effect of Spatial Smoothness. To examine how spatial smoothness affects solution performance, we vary the spatial smoothness threshold over $\Delta^b \in \{0.5, 1.0, 2.0, +\infty\}$, where larger values permit greater spatial differentiation. We measure spatial bonus variation as the mean of $|b_i - b_j|$ over neighboring zone pairs. Table 5 shows that relaxing the threshold initially improves performance, but the gains diminish quickly. When $\kappa = 0.50$, increasing Δ^b from 0.5 to 2.0 raises the objective improvement from 28.61% to 31.99%, whereas removing the constraint increases bonus variation without further improvement. Similar patterns arise when $\kappa = 1.00$ and 1.10, suggesting that a moderate threshold preserves most of the targeting value while avoiding sharp bonus differences across neighboring zones. Corollary 2 explains how smoothness changes the first-order targeting rule. Edges with positive smoothness multipliers can sustain opposing local marginal-value pressures, but those pressures must balance over the associated component unless a bonus bound absorbs them. The numerical results measure the attained value of relaxing the threshold, whereas the corollary identifies the spatial mechanism through which this relaxation can matter.

6.6. Anytime Performance

To assess performance within the 120-second operating window, we compare the incumbent trajectories of ISA and NOMAD across the 26 unique generated instances used in the supply–demand regime and scalability experiments. For method m, let $Z_m(t)$ denote the objective value of its

Table 5 Sensitivity analysis with respect to the spatial smoothness threshold.

κ	Δ^b	Obj.	Obj. Imp. (%)	Match rate (%)	Mean $\lvert b_i - b_j \rvert$	ROI (%)	Time
0.50	0.5	13895	28.61	56.94	0.221	618.12	33.15
	1.0	14183	31.28	57.74	0.249	675.83	30.84
	2.0	14261	31.99	58.02	0.262	691.31	27.27
	$+\infty$	13936	28.98	56.86	0.533	626.31	37.62
1.00	0.5	18336	14.57	98.12	0.210	466.43	27.10
	1.0	18631	16.42	99.30	0.251	525.46	29.04
	2.0	18714	16.94	99.76	0.232	542.15	37.98
	$+\infty$	18713	16.93	99.76	0.232	541.99	35.89
1.10	0.5	18568	12.07	100.00	0.212	400.06	26.73
	1.0	18842	13.73	100.00	0.266	454.89	29.97
	2.0	18929	14.25	100.00	0.240	472.21	31.31
	$+\infty$	18929	14.25	100.00	0.241	472.30	36.10

latest feasible incumbent available by time t, with the incumbent carried forward if the method terminates early. We measure ISA's objective lead over NOMAD at time t as $L(t) := [Z_{\text{ISA}}(t) - Z_{\text{NOMAD}}(t)]/|Z_{\text{NOMAD}}(t)| \times 100\%$. We also compare how quickly the two methods reach a common objective target. Let $Z^{120}_{\max} := \max\{Z_{\text{ISA}}(120), Z_{\text{NOMAD}}(120)\}$, and define $Z_\alpha := Z^{\text{no}} + \alpha(Z^{120}_{\max} - Z^{\text{no}})$ for $\alpha \in \{0.90, 0.95, 0.99\}$, where Z^{no} is the no-bonus objective. Thus, the target represents a fraction α of the improvement from the no-bonus objective to the better 120-second incumbent. The target time and number of matching-LP solves are recorded when a method first obtains an incumbent satisfying $Z_m(t) \geq Z_\alpha$. These targets measure progress relative to the observed incumbents and are distinct from the solver-certified upper bound.

Figure 3 summarizes the incumbent trajectories and attainment of the common 95% target. ISA reaches this target in a median of 6.36 seconds, whereas NOMAD remains below it throughout the operating window on all 26 instances. By 60 seconds, ISA's incumbent also exceeds the best completed non-ISA feasible solution on every instance. Detailed results, including matching-LP solve counts, are reported in Section EC.6.2 of the e-companion.

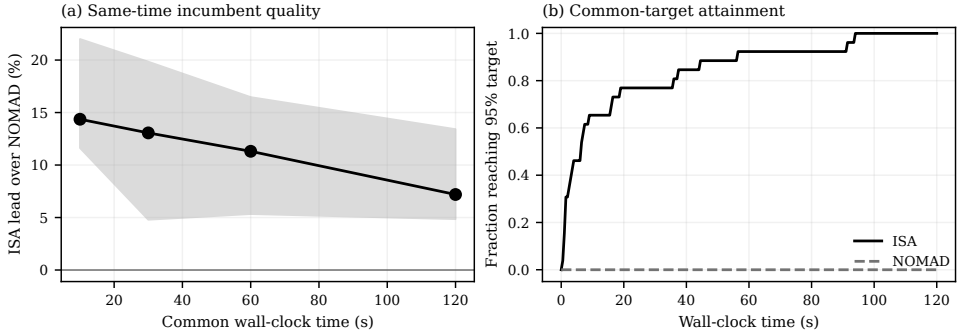


Figure 3 Anytime performance on 26 unique generated instances.

6.7. Scalability

To evaluate the scalability of ISA, we conduct experiments by varying the number of zones over $|\mathcal{Z}| \in \{50, 100, 150, 200, 250, 300\}$ under moderate-shortage, balanced, and oversupplied regimes,

while holding all other parameters fixed. Table 6 shows that ISA attains the highest feasible objective among the evaluated methods in every instance, improving on the strongest non-ISA feasible benchmark by 0.36%–18.74%. At 300 zones, these improvements remain 9.00%–15.19% across the three supply regimes. ISA returns solutions for instances with up to 200 zones in approximately 70 seconds and retains the best incumbent accepted by the nominal 120-second deadline for the 250- and 300-zone instances.

Table 6 Scalability across problem sizes.

$\lvert\mathcal{Z}\rvert$	κ	Conic		$\mathbf{B}^0$		$\mathbf{H}_1$		$\mathbf{H}_2$		NOMAD		UB		ISA	
		Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time
50	0.50	820	0.35	7052	0.01	7374	0.02	7258	0.03	7950	23.37	12540	0.34	**8393** ↑5.58%	2.44
	1.00	1151	0.13	13490	0.01	13495	0.02	13646	0.03	14087	23.84	15409	0.12	**14186** ↑0.70%	13.14
	1.10	1151	0.14	14644	0.01	14523	0.02	14734	0.03	14881	36.06	15409	0.13	**14934** ↑0.36%	2.26
100	0.50	2957	0.74	6628	0.03	4909	0.09	6854	0.11	8819	101.92	12892	0.71	**9367** ↑6.21%	15.28
	1.00	1951	0.69	10837	0.03	6316	0.09	10929	0.11	12477	96.63	15190	0.66	**12850** ↑2.99%	9.43
	1.10	1856	0.66	11386	0.03	6373	0.09	11386	0.06	12747	113.48	15310	0.63	**13030** ↑2.22%	9.53
150	0.50	3470	1.41	10804	0.07	9038	0.18	11111	0.25	12179	119.76	19403	1.33	**14183** ↑16.46%	30.84
	1.00	2286	1.41	16004	0.09	13145	0.19	16004	0.16	17700	119.77	22076	1.35	**18631** ↑5.26%	29.04
	1.10	2143	1.39	16568	0.09	13814	0.17	16568	0.18	17951	119.65	22251	1.30	**18842** ↑4.96%	29.97
200	0.50	2410	3.13	14676	0.16	12745	0.34	15086	0.46	15488	119.77	26239	2.96	**18392** ↑18.74%	70.34
	1.00	1667	2.68	19373	0.16	17841	0.36	19373	0.30	21857	119.77	28697	2.51	**22295** ↑2.00%	69.77
	1.10	1578	2.79	19792	0.18	18510	0.36	19792	0.30	22209	119.77	28994	2.61	**22613** ↑1.82%	70.19
250	0.50	2820	4.73	17455	0.28	16799	0.54	17845	0.78	18303	119.77	29753	4.47	**21166** ↑15.64%	120.09
	1.00	2009	4.30	21319	0.28	20193	0.58	21319	0.58	21319	119.77	32432	4.06	**24530** ↑15.06%	120.02
	1.10	1959	4.32	21741	0.28	20656	0.60	21741	0.54	21760	119.77	32758	4.07	**24885** ↑14.36%	120.04
300	0.50	2623	7.41	21825	0.41	20427	0.80	22454	1.19	22787	118.89	40192	7.05	**26249** ↑15.19%	120.15
	1.00	1714	6.57	31032	0.39	29925	0.84	31032	0.80	31032	119.77	44870	6.20	**34207** ↑10.23%	120.11
	1.10	1636	6.93	31862	0.40	30910	0.85	31862	0.81	31862	119.77	45306	6.56	**34731** ↑9.00%	120.13

6.8. New York City Study

The generated instances allow us to evaluate the model under controlled operating conditions. We next complement these experiments with a data-based case study to assess ISA in a realistic large-scale setting. Specifically, we use 2015 New York City Yellow Taxi trip records released by the Taxi and Limousine Commission (New York City Taxi and Limousine Commission 2026). The full preprocessing region comprises 2,350 Geohash-6 cells; each daily instance retains cells appearing as pickup or dropoff locations, leaving approximately 1,400 active zones. Demand equals the number of pickups in a two-minute target window, and initial dispatchable supply is approximated by the number of dropoffs in the preceding minute. Latent supply is set to 0.3 times total demand and allocated across zones in proportion to demand. The pre-bonus matching value is $p_{ij} = 15 + 10\exp(-d_{ij}/3)$, where d_{ij} is the Haversine distance in kilometers.

For a daily instance with $n = |\mathcal{Z}|$, let $m = \lceil \sqrt{n} \rceil$. We partition the latitude–longitude bounding box of the observed zone centers into an $m \times m$ equal-width grid and connect zones whose cells coincide or are horizontally, vertically, or diagonally adjacent. For these instances, the 3×3 cell

stencil defines the sparse graph used for driver relocation, spatial smoothness, and matching. The study combines taxi-based geography, demand, and a dispatchable-supply proxy with specified latent supply, matching values, and driver response parameters.

We select three dates for each of the intervals 07:00–07:02, 10:00–10:02, and 17:00–17:02 to represent morning peak, off-peak, and evening peak conditions. The nine instances contain 1,396–1,486 active zones and preserve temporal variation and heterogeneous spatial imbalances. Table 7 summarizes these instances and uses M, O, and E to denote the three market conditions, where $D^{\text{tot}} = \sum_{i\in\mathcal{Z}} d_i$, $S^{\text{tot}} = \sum_{i\in\mathcal{Z}} s_i^{\text{On}}$, and $D^{\text{gap}} = \sum_{i\in\mathcal{Z}} \max\{d_i - s_i^{\text{On}}, 0\}$ denote total rider demand, the preceding-minute dropoff proxy for dispatchable supply, and the total local shortage gap, respectively. We compare ISA with the benchmarks in Section 6.2 using a total bonus budget of

Table 7 Details of the New York City instances.

Type	ID	Date	Period	$\lvert\mathcal{Z}\rvert$	D^{tot}	S^{tot}	D^{gap}	κ
Morning peak	M1	2015-01-20	07:00–07:02	1417	521	278	338	0.53
	M2	2015-01-22		1486	574	274	384	0.48
	M3	2015-01-29		1459	570	267	377	0.47
Off-peak	O1	2015-01-08	10:00–10:02	1485	680	380	363	0.56
	O2	2015-01-18		1464	539	292	286	0.54
	O3	2015-01-24		1418	550	314	281	0.57
Evening peak	E1	2015-01-03	17:00–17:02	1396	648	343	350	0.53
	E2	2015-01-07		1428	614	323	341	0.53
	E3	2015-01-17		1421	680	386	359	0.57

$B = 1500$ for every instance; both one-pass heuristics complete within the 120-second optimization time limit. Table 8 shows that, under the nominal 120-second optimization deadline, ISA attains the best objective in all nine instances and improves upon the strongest non-ISA feasible benchmark by 2.39%–8.43%. NOMAD retains its $\mathbf{H}_1$ starting solution throughout the run, whereas the Conic benchmark provides a recovered feasible solution and the conic outer relaxation provides a solver-certified upper bound for every instance.

Table 8 Performance comparison in the New York City study.

Type	ID	Conic		$\mathbf{B}^0$		$\mathbf{H}_1$		$\mathbf{H}_2$		NOMAD		UB		ISA	
		Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time	Obj.	Time
Morning peak	M1	5796	84.82	6576	0.20	8224	1.30	6730	0.67	8224	119.77	10495	84.62	**8706** ↑5.86%	120.36
	M2	5251	71.77	6548	0.20	8276	1.32	6727	0.73	8276	119.78	10828	71.55	**8974** ↑8.43%	120.19
	M3	5107	61.38	6466	0.20	8371	1.31	6643	0.72	8371	119.77	10688	61.16	**8881** ↑6.10%	120.11
Off-peak	O1	5284	75.50	9157	0.22	11348	1.35	9354	0.73	11348	119.78	14322	75.28	**11871** ↑4.61%	120.18
	O2	5009	66.78	6994	0.20	8777	1.34	7154	0.70	8777	119.78	11050	66.58	**9182** ↑4.62%	120.44
	O3	5781	119.50	7616	0.21	9555	1.34	7771	0.71	9555	119.77	11798	119.29	**9791** ↑2.47%	120.79
Evening peak	E1	5284	69.66	8191	0.18	10479	1.21	8383	0.67	10479	119.78	13113	69.47	**10829** ↑3.34%	120.14
	E2	5095	55.70	7760	0.20	10015	1.34	7941	0.71	10015	119.78	12359	55.48	**10254** ↑2.39%	120.96
	E3	5231	84.34	8969	0.20	11241	1.32	9165	0.73	11241	119.78	14193	84.12	**11716** ↑4.23%	120.15

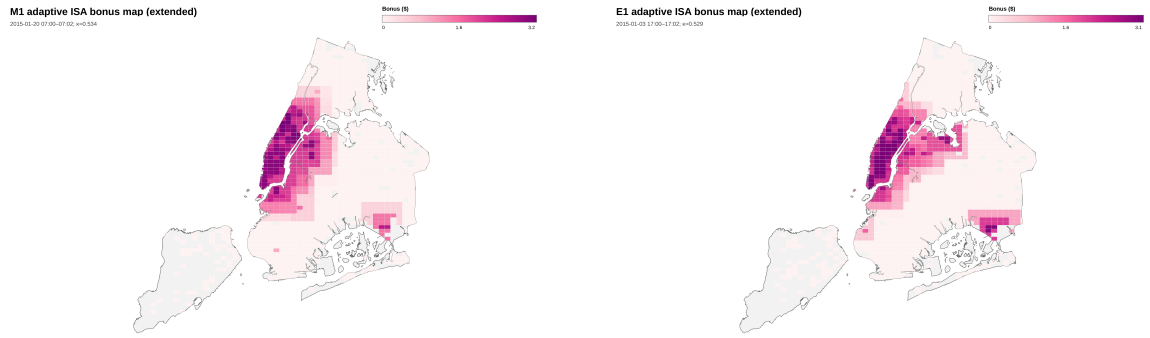
To assess objective stability beyond the 120-second operating window, we compare the cutoff incumbents with those from extended runs. Table 9 shows that the extended runs yield additional objective gains of up to 0.16%, indicating limited improvement beyond the operating window for the tested instances. The columns Obj. (120 s) and Obj. (Ext.) report the cutoff and extended-run

objectives, respectively, and Ext. gain reports their relative difference. Cum. time denotes the total wall-clock time across both phases.

Table 9 Comparison of 120-second and extended ISA incumbents in the New York City study.

Type	ID	Obj. (120 s)	Obj. (Ext.)	Ext. gain (%)	Cum. time
Morning peak	M1	8706	8707	0.02	845.32
	M2	8974	8989	0.16	928.99
	M3	8881	8885	0.04	935.27
Off-peak	O1	11871	11881	0.09	924.99
	O2	9182	9184	0.02	901.76
	O3	9791	9791	0.00	1920.88
Evening peak	E1	10829	10831	0.02	874.05
	E2	10254	10254	0.00	871.44
	E3	11716	11718	0.01	915.90

Beyond the objective comparisons, the optimized maps show how ISA allocates bonuses across space. Figure 4 shows the ISA maps for the New York City M1 and E1 instances. Within each panel, darker shading denotes a higher driver bonus, whereas blank or very light zones correspond to zero or low bonuses. In Figure 4a, high bonuses are mainly concentrated along the Manhattan street grid, especially in central and lower Manhattan. In Figure 4b, high-bonus zones appear across Manhattan and extend into nearby surrounding areas, reflecting a more spatially dispersed evening supply–demand imbalance.



(a) M1 (b) E1

Figure 4 Optimized driver surge maps for two New York City instances.

7. Conclusion

This paper studies how ride-hailing platforms can design driver surge maps in real time. We formulate the DSMDP to optimize zone-level bonuses while anticipating voluntary driver relocation and participation and the resulting downstream matching value. Exploiting the linear-program structure of the matching problem, ISA samples dual sensitivities across nearby matching regimes to guide feasible bonus-map updates. Theoretical analysis provides finite-scale stationarity certificates and identifies sufficient conditions for constrained Clarke stationarity of sampled limits. A conic outer relaxation provides a valid upper bound for evaluating feasible solutions.

The computational results show that ISA attains the highest net operating profit among the evaluated methods, improving on the strongest non-ISA feasible benchmark by up to 18.74% on the generated instances and by 2.39%–8.43% in the New York City study. The results suggest that platforms should direct bonuses toward responsive supply whose downstream matching value justifies the associated payments, rather than relying on local shortage alone. Intervention generates the greatest incremental value under supply shortage, but additional supply can remain valuable even when all demand is served by supporting higher-value matches. Holding activation responsiveness fixed, a small bonus has the greatest effect on latent-driver activation when drivers are equally likely to participate or remain inactive without incentives. The value of activating these drivers depends on the matching opportunities available to the platform. Higher bonuses increase payments on existing matches and can also incur a budget opportunity cost when the bonus budget is binding. The budget study shows diminishing returns to additional spending. Finally, moderate spatial smoothness preserves most of the gains from spatial targeting, suggesting that effective bonus maps need not display sharp differences between neighboring zones.

We conclude with several directions for extending the current framework. The analysis is based on a single-period model, pre-estimated behavioral-response parameters, and a platform net operating profit objective. Future research could extend the framework to dynamic settings in which current bonuses affect future supply and matching opportunities, incorporate heterogeneous and strategic driver behavior such as multi-homing and trip acceptance, and develop robust or data-driven methods for uncertain demand, driver responses, and real-time supply observations. Extensions could also examine how service quality, driver earnings, and spatial equity alter the design of driver surge maps.

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E-Companion to Driver Surge Maps for Supply Rebalancing with Downstream Matching

EC.1. Calibration of Driver Response Parameters

In deployment, the behavioral-response parameters introduced in the mathematical formulation can be estimated from historical data before optimization and treated as fixed inputs when solving the DSMDP. Specifically, the platform can construct a large sample of driver transitions between consecutive decision periods using historical platform records. For each dispatchable driver, the data record the current zone, the bonuses displayed in nearby reachable zones, and the location or dispatch-availability state in the next period. Transitions associated with dispatched trips can be excluded to isolate voluntary responses.

For observation n, let i_n denote the current zone, $\boldsymbol{b}_n$ the displayed bonus vector, and y_n the observed next-period choice, which is either a reachable zone $j \in \mathcal{Z}_{i_n}$ or the unavailable state. The parameters $\boldsymbol{\beta}$, $\boldsymbol{\delta}^{\mathrm{Off}}$, and ζ^{Off} can then be estimated by maximizing the multinomial logit (MNL) log-likelihood $\sum_n \left[\sum_{j\in\mathcal{Z}_{i_n}} \mathbb{1}_{\{y_n=j\}} \log P_{i_n j}(\boldsymbol{b}_n) + \mathbb{1}_{\{y_n=\mathrm{Off}\}} \log P^{\mathrm{Off}}_{i_n}(\boldsymbol{b}_n)\right]$ subject to $\zeta^{\mathrm{Off}} \geq 0$. The likelihood is expressed in terms of the normalized utilities $V_{ij} = \beta^{(1)} b_j + \beta^{(3)} d_{ij}$ and $V_i^{\mathrm{Off}} = \delta_i^{\mathrm{Off}} - \zeta^{\mathrm{Off}} b_i$, obtained by subtracting the common dispatchable-state utility term $\beta^{(0)} + \beta^{(2)} b_i$. The normalization yields $\delta_i^{\mathrm{Off}} = \alpha_i^{\mathrm{Off}} - \beta^{(0)}$ and $\zeta^{\mathrm{Off}} = \beta^{(2)} + \gamma^{\mathrm{Off}}$, allowing the identified coefficient combinations to be estimated directly while preserving the original choice probabilities.

Similarly, for latent drivers, the platform observes their current zone, local bonus, and whether they enter the dispatchable pool in the next period. Let $a_n = 1$ if observation n records activation into the dispatchable pool and 0 otherwise. The parameters $\boldsymbol{\alpha}^{\mathrm{On}}$ and γ^{On} can be estimated using the binary logit likelihood $\sum_n \left[a_n \log P^{\mathrm{On}}_{i_n}(\boldsymbol{b}_n) + (1-a_n)\log\left(1 - P^{\mathrm{On}}_{i_n}(\boldsymbol{b}_n)\right)\right]$. The estimated parameters may be updated periodically using a rolling historical sample and are held fixed during each real-time decision period.

EC.2. Derivatives of Driver Response Probabilities

The utility derivatives used by the Lagrangian-based gradient update are

$$\frac{\partial V_{ij}}{\partial b_m} = \beta^{(1)} \mathbb{1}_{\{j=m\}}, \qquad \frac{\partial V_i^{\mathrm{Off}}}{\partial b_m} = -\zeta^{\mathrm{Off}} \mathbb{1}_{\{i=m\}}, \qquad i \in \mathcal{Z},\ j \in \mathcal{Z}_i,\ m \in \mathcal{Z}. \tag{EC.1}$$

Substituting the utility derivatives in Eq. (EC.1) into the MNL derivative gives, for $i, m \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$,

$$\frac{\partial P_{ij}(\boldsymbol{b})}{\partial b_m} = P_{ij}(\boldsymbol{b})\left[\beta^{(1)}\left(\mathbb{1}_{\{j=m\}} - \mathbb{1}_{\{m\in\mathcal{Z}_i\}} P_{im}(\boldsymbol{b})\right) + \zeta^{\mathrm{Off}} P_i^{\mathrm{Off}}(\boldsymbol{b}) \mathbb{1}_{\{i=m\}}\right]. \tag{EC.2}$$

The parameter ζ^{Off} represents the identified relative effect of the origin-zone bonus on the dispatchable-versus-unavailable margin. The derivative of the unavailability probability is

$$\frac{\partial P_i^{\mathrm{Off}}(\boldsymbol{b})}{\partial b_m} = -P_i^{\mathrm{Off}}(\boldsymbol{b})\left[\beta^{(1)} \mathbb{1}_{\{m\in\mathcal{Z}_i\}} P_{im}(\boldsymbol{b}) + \zeta^{\mathrm{Off}}\left(1 - P_i^{\mathrm{Off}}(\boldsymbol{b})\right) \mathbb{1}_{\{i=m\}}\right], \qquad i, m \in \mathcal{Z}. \tag{EC.3}$$

The derivative of the latent-driver activation probability is

$$\frac{\partial P_i^{\mathrm{On}}(\boldsymbol{b})}{\partial b_m} = \gamma^{\mathrm{On}} P_i^{\mathrm{On}}(\boldsymbol{b}) \left(1 - P_i^{\mathrm{On}}(\boldsymbol{b})\right) \mathbb{1}_{\{i=m\}}, \qquad \forall i, m \in \mathcal{Z}. \tag{EC.4}$$

Combining Eqs. (EC.2), (EC.3), and (EC.4), differentiation of the supply-balance expression yields

$$\begin{aligned} \frac{\partial s_i(\boldsymbol{b})}{\partial b_m} = & \sum_{j \in \mathcal{Z}_i \setminus \{i\}} s_j^{\mathrm{On}} \frac{\partial P_{ji}(\boldsymbol{b})}{\partial b_m} \\ & - s_i^{\mathrm{On}} \left[\sum_{j \in \mathcal{Z}_i \setminus \{i\}} \frac{\partial P_{ij}(\boldsymbol{b})}{\partial b_m} + \frac{\partial P_i^{\mathrm{Off}}(\boldsymbol{b})}{\partial b_m} \right] \\ & + s_i^{\mathrm{Off}} \frac{\partial P_i^{\mathrm{On}}(\boldsymbol{b})}{\partial b_m}, \qquad \forall i, m \in \mathcal{Z}. \end{aligned} \tag{EC.5}$$

EC.2.1. Derivation of the Marginal Bonus Value

The supply sensitivity in Eq. (EC.5) enters the Lagrangian derivative below. Holding the selected optimal primal-dual solution fixed, differentiating the Lagrangian of Objective (3a) and Constraints (3b)–(3d) with multipliers $(\boldsymbol{\pi}, \boldsymbol{\eta}, \lambda)$ gives Eq. (4); the direct payout and budget derivatives combine as $-(1+\lambda)q_m$.

Proof of Proposition 1. Given $\sum_{j \in \mathcal{Z}_i} P_{ij}(\boldsymbol{b}) + P_i^{\mathrm{Off}}(\boldsymbol{b}) = 1$, the supply-balance equation can be written as

$$s_i(\boldsymbol{b}) = \sum_{j:\, i \in \mathcal{Z}_j} s_j^{\mathrm{On}} P_{ji}(\boldsymbol{b}) + s_i^{\mathrm{Off}} P_i^{\mathrm{On}}(\boldsymbol{b}). \tag{EC.6}$$

Using $\bar{\eta}_i$ as defined before Proposition 1, Eq. (EC.2) gives Eq. (5). Differentiating Eq. (EC.6), weighting the result by the supply-constraint dual multipliers, and applying Eqs. (5) and (EC.4) therefore yields

$$\begin{aligned} \sum_{j \in \mathcal{Z}} \eta_j \frac{\partial s_j(\boldsymbol{b})}{\partial b_m} = & \beta^{(1)} \sum_{i:\, m \in \mathcal{Z}_i} s_i^{\mathrm{On}} P_{im}(\boldsymbol{b}) \left(\eta_m - \bar{\eta}_i\right) \\ & + \zeta^{\mathrm{Off}} s_m^{\mathrm{On}} P_m^{\mathrm{Off}}(\boldsymbol{b}) \bar{\eta}_m \\ & + \gamma^{\mathrm{On}} s_m^{\mathrm{Off}} P_m^{\mathrm{On}}(\boldsymbol{b}) \left(1 - P_m^{\mathrm{On}}(\boldsymbol{b})\right) \eta_m. \end{aligned} \tag{EC.7}$$

Substituting Eq. (EC.7) into Eq. (4) proves Eq. (7). □

EC.2.2. Proofs of the Intervention and Spatial Balance Results

Proof of Corollary 1. At the no-bonus map, realized bonus spending is zero and the budget constraint is strictly slack because $B > 0$. Complementary slackness therefore gives $\lambda^{\mathrm{no}} = 0$. Differentiability of Z at $\mathbf{0}$ and Proposition 1 then give Eq. (8). Moreover, $b_i \boldsymbol{e}_i \in \mathcal{B}$ for every sufficiently small $b_i > 0$ because $b^{\max} > 0$ and $\Delta^b > 0$. A positive one-sided derivative consequently yields $Z(b_i \boldsymbol{e}_i) > Z(\mathbf{0})$ for every sufficiently small positive b_i.

At $\mathbf{0}$, the tangent cone of $\mathcal{B}$ is the nonnegative orthant. The first-order stationarity condition is therefore equivalent to $g_i^{\mathrm{no}} \le 0$ for every $i \in \mathcal{Z}$. If $g_i^{\mathrm{no}} < 0$ for every $i \in \mathcal{Z}$, differentiability at $\mathbf{0}$ gives

$$Z(\boldsymbol{b}) - Z(\mathbf{0}) = (\boldsymbol{g}^{\mathrm{no}})^{\top} \boldsymbol{b} + o(\|\boldsymbol{b}\|) < 0$$

for every sufficiently small nonzero feasible $\boldsymbol{b}$, since $\boldsymbol{b} \geq \mathbf{0}$. Thus, the no-bonus map is a strict local maximizer. □

Derivation of Eq. (10). In the single-zone setting, Eq. (9) is the reduced objective on a right neighborhood of zero. The right derivative of $Z_i(b_i)$ at zero, denoted by $Z'_{i,+}(0)$, is

$$Z'_{i,+}(0) = p_{ii} s_i^{\mathrm{Off}} \gamma^{\mathrm{On}} P_i^{\mathrm{On,no}} \left(1 - P_i^{\mathrm{On,no}}\right) - \left(s_i^{\mathrm{On}} + s_i^{\mathrm{Off}} P_i^{\mathrm{On,no}}\right).$$

Rearranging the condition $Z'_{i,+}(0) > 0$ yields Eq. (10). □

Proof of Proposition 2 and Corollary 2. Since $\boldsymbol{g}^\star = \nabla Z(\boldsymbol{b}^\star)$, local optimality over the convex set $\mathcal{B}$ implies

$$(\boldsymbol{g}^\star)^\top (\boldsymbol{b} - \boldsymbol{b}^\star) \leq 0, \qquad \forall \boldsymbol{b} \in \mathcal{B}, \tag{EC.8}$$

The variational inequality in Eq. (EC.8) gives $\boldsymbol{g}^\star \in \mathcal{N}_{\mathcal{B}}(\boldsymbol{b}^\star)$. The normal cone of the polyhedral set in Eq. (11) is the nonnegative hull of its active outward normals, which yields Eq. (12).

For an interior zone with no incident spatial multiplier, every term on the right-hand side of the corresponding coordinate in Eq. (12) is zero. Equation (13) then follows from Eq. (7). For a component C, every edge outside $\mathcal{E}_\rho$ has zero multiplier, and every edge in $\mathcal{E}_\rho$ has both endpoints in the same component. Let $\boldsymbol{\rho}^\pm := (\rho_e^\pm)_{e \in \mathcal{E}^b}$. Summing the coordinates of $\boldsymbol{D}^\top(\boldsymbol{\rho}^+ - \boldsymbol{\rho}^-)$ over C gives zero because each incidence row contributes once with sign $+1$ and once with sign -1. Equations (14) and (15) follow. □

EC.3. Adaptive Constrained Gradient Sampling

Following the gradient-sampling principle (Burke et al. 2002, Curtis and Overton 2012), ISA samples sensitivities from nearby matching branches to address transitions between locally smooth regimes. We specify the sampling, null-step, scale-refinement, and stopping procedures omitted from Algorithm 1. The construction links the sampled sensitivity bundle to feasible ascent, stationarity assessment, and adaptive scale refinement.

EC.3.1. Sampled Constrained Direction

At the center $\boldsymbol{b}^k \in \mathcal{B}$, the sampling radius ϵ_k, stationarity scale ν_k, and sample pool may be updated over several attempts while the center remains fixed, with all existing evaluations reused. Define $\mathcal{I}_k := \{0\} \cup \mathcal{A}_k$, where $a = 0$ denotes the evaluation at the center and $\mathcal{A}_k$ indexes the ambient samples. Set $\boldsymbol{y}^{k,0} := \boldsymbol{b}^k$ and draw each $\boldsymbol{y}^{k,a}$, $a \in \mathcal{A}_k$, uniformly from $\mathbb{B}_{\epsilon_k}(\boldsymbol{b}^k) := \{\boldsymbol{y} : \|\boldsymbol{y} - \boldsymbol{b}^k\|_2 \leq \epsilon_k\}$.

Suppose that Z admits the locally Lipschitz extension used in Propositions 3 and 4. The ambient samples are then differentiability points with probability one whenever their sampling balls lie within this neighborhood. We assume that, at each ambient sample $\boldsymbol{y}^{k,a}$, the sensitivity computed from Eq. (4) equals $\nabla Z(\boldsymbol{y}^{k,a})$, so that $\boldsymbol{g}^{k,a} = \nabla Z(\boldsymbol{y}^{k,a})$ for each $a \in \mathcal{A}_k$. The center sensitivity $\boldsymbol{g}^{k,0}$ is the sensitivity of the selected center branch and is retained in the working bundle even when

differentiability at the center has not been established. Let $\boldsymbol{G}_k := [\boldsymbol{g}^{k,a}]_{a\in\mathcal{I}_k}$ denote this working bundle.

The constraints defining $\mathcal{B}$ enter through $\boldsymbol{N}_k$, while the realized bonus-spending budget enters each sampled sensitivity through the multiplier λ in Eq. (4). ISA then solves

$$(\boldsymbol{\omega}^k, \boldsymbol{\mu}^k) \in \arg\min_{\substack{\boldsymbol{\omega}\geq\mathbf{0},\ \mathbf{1}^\top\boldsymbol{\omega}=1 \\ \boldsymbol{\mu}\geq\mathbf{0}}} \frac{1}{2}\left\|\boldsymbol{G}_k\boldsymbol{\omega} - \boldsymbol{N}_k\boldsymbol{\mu}\right\|_2^2 \tag{EC.9}$$

$$\bar{\boldsymbol{g}}^k := \boldsymbol{G}_k\boldsymbol{\omega}^k, \qquad \bar{\boldsymbol{n}}^k := \boldsymbol{N}_k\boldsymbol{\mu}^k, \qquad \boldsymbol{r}^k := \bar{\boldsymbol{g}}^k - \bar{\boldsymbol{n}}^k, \qquad \widehat{R}_k := \|\boldsymbol{r}^k\|_2. \tag{EC.10}$$

The QP optimality conditions imply $\boldsymbol{N}_k^\top\boldsymbol{r}^k \leq \mathbf{0}$ and $\min_{a\in\mathcal{I}_k}(\boldsymbol{g}^{k,a})^\top\boldsymbol{r}^k \geq \|\boldsymbol{r}^k\|_2^2$. Hence, $\boldsymbol{r}^k$ is a feasible common ascent direction for the sampled sensitivities whenever $\widehat{R}_k > 0$. The projected line search below tests this direction while preserving the bonus-feasible set $\mathcal{B}$.

EC.3.2. Projected Bundle Line Search and Null Steps

Fix an initial step $\bar{\tau} > 0$, contraction factor $\delta \in (0,1)$, Armijo sufficient-increase parameter $\chi \in (0,1)$, maximum number of trials L_k, and minimum trial step $\tau_{\min} > 0$. For $\ell = 0, \ldots, L_k - 1$, set $\tau_{k,\ell} := \bar{\tau}\delta^\ell$, $\boldsymbol{h}^{k,\ell} := \Pi_{\mathcal{B}}(\boldsymbol{b}^k + \tau_{k,\ell}\boldsymbol{r}^k) - \boldsymbol{b}^k$, and $\Gamma_k(\boldsymbol{h}^{k,\ell}) := \min_{a\in\mathcal{I}_k}(\boldsymbol{g}^{k,a})^\top\boldsymbol{h}^{k,\ell}$. To avoid vanishingly small steps, trials with step sizes below $\tau_{\min}$ are not evaluated. A trial is accepted if $\Gamma_k(\boldsymbol{h}^{k,\ell}) > 0$ and $Z(\boldsymbol{b}^k + \boldsymbol{h}^{k,\ell}) \geq Z(\boldsymbol{b}^k) + \chi\Gamma_k(\boldsymbol{h}^{k,\ell})$.

An accepted trial sets $\boldsymbol{b}^{k+1} := \boldsymbol{b}^k + \boldsymbol{h}^{k,\ell}$ and resets the sample pool around the new center. If no trial is accepted, the center is retained and a batch of new ambient evaluations is added. Once the pool reaches its capacity, ambient sensitivities with the smallest QP weights are removed, while the center sensitivity is retained. When a time, accepted-step, or sampling-attempt limit is reached, ISA returns the best center found. The working-bundle residual $\widehat{R}_k$ and accumulated ambient samples feed into the scale-refinement and stationarity assessment described next.

EC.3.3. Adaptive Sampling Radius and Stationarity Assessment

ISA uses an attempt window W and an objective-stability threshold $\eta_Z > 0$. Let $Z^{[u]}$ denote the objective value at the center for target-scale attempt u. Once $W+1$ observations are available, define $\Delta_Z^{[u]} := \max\{0, Z^{[u]} - Z^{[u-W]}\}/\max\{1, |Z^{[u]}|\}$. Objective stability holds when $\Delta_Z^{[u]} \leq \eta_Z$.

When $\widehat{R}_k \leq \nu_k$ and either the current scales are coarse or the target-scale objective-stability test is satisfied, ISA enriches the ambient pool to contain at least $|\mathcal{Z}| + 1$ points. It then re-solves Eq. (EC.9) using only $\boldsymbol{G}_k^{\mathrm{amb}} := [\boldsymbol{g}^{k,a}]_{a\in\mathcal{A}_k}$, thereby omitting the deterministic center sensitivity $\boldsymbol{g}^{k,0}$. Let $\widehat{R}_k^{\mathrm{amb}}$ and $\boldsymbol{r}^{k,\mathrm{amb}}$ denote the resulting residual and direction, respectively. At the target scales, this ambient assessment is deferred until objective stability holds; earlier attempts continue with the working bundle $\boldsymbol{G}_k$. Proposition 3 shows that an ambient residual $\widehat{R}_k^{\mathrm{amb}} \leq \nu_k$, paired with radius ϵ_k, gives an (ϵ_k, ν_k)-stationarity certificate.

Scale refinement or termination occurs only when $\widehat{R}_k^{\mathrm{amb}} \le \nu_k$. If this condition holds at $(\epsilon_{\mathrm{tar}}, \nu_{\mathrm{tar}})$, ISA returns the current center. At a coarser scale, it sets $\epsilon_k \leftarrow \max\{\epsilon_{\mathrm{tar}}, \kappa_\epsilon \epsilon_k\}$ and $\nu_k \leftarrow \max\{\nu_{\mathrm{tar}}, \kappa_\nu \nu_k\}$, where $\kappa_\epsilon, \kappa_\nu \in (0,1)$, and rebuilds the pool at the same center. If $\widehat{R}_k^{\mathrm{amb}} > \nu_k$, the subsequent line search uses $\boldsymbol{r}^{k,\mathrm{amb}}$ and the ambient bundle $\boldsymbol{G}_k^{\mathrm{amb}}$. For this line search, replace $\boldsymbol{r}^k$ by $\boldsymbol{r}^{k,\mathrm{amb}}$ and $\mathcal{I}_k$ by $\mathcal{A}_k$ in the line-search formulas. Proposition 4 establishes constrained Clarke stationarity along convergent ambient assessments satisfying $\epsilon_\ell \downarrow 0$ and $\widehat{R}_\ell^{\mathrm{amb}} \to 0$.

EC.3.4. Smooth Projected Gradient Specialization

For the supplementary smooth-case analysis, we consider a specialization that uses only the center gradient. Under the stable optimal-basis condition in Section 4.1, the Lagrangian sensitivity equals $\nabla Z(\boldsymbol{b})$.

Fix an initial step $\bar{\tau} > 0$, a contraction factor $\delta \in (0,1)$, and an Armijo parameter $\chi \in (0,1)$. For $\tau_{k,\ell} := \bar{\tau}\delta^\ell$, $\ell = 0, 1, \ldots$, define

$$\boldsymbol{h}_{k,\ell} := \Pi_{\mathcal{B}}\big(\boldsymbol{b}^k + \tau_{k,\ell} \nabla Z(\boldsymbol{b}^k)\big) - \boldsymbol{b}^k.$$

The procedure accepts the first trial satisfying

$$Z(\boldsymbol{b}^k + \boldsymbol{h}_{k,\ell}) \ge Z(\boldsymbol{b}^k) + \chi \nabla Z(\boldsymbol{b}^k)^\top \boldsymbol{h}_{k,\ell},$$

and sets $\boldsymbol{b}^{k+1} := \boldsymbol{b}^k + \boldsymbol{h}_{k,\ell}$. The convergence and accepted-step complexity of this specialization are established in Section EC.4.4 of the e-companion.

EC.4. Proofs of Theoretical Results

The proofs use the normal-cone geometry of $\mathcal{B}$ and parametric-LP sensitivity to connect the sampled directions to stationarity. We establish continuity and the finite-scale certificate before analyzing smooth accepted steps and shrinking-scale Clarke limits.

EC.4.1. Proof of Lemma 1

Proof. Let $\mathcal{X}(\boldsymbol{b})$ denote the feasible set of DSMDP-M, and define $q_i(\boldsymbol{x}) := \sum_{j \in \mathcal{Z}} x_{ij}$ and $\boldsymbol{q}(\boldsymbol{x}) := (q_i(\boldsymbol{x}))_{i \in \mathcal{Z}}$. The zero matching is feasible for every $\boldsymbol{b}$, and the demand constraints provide the uniform bound $\sum_{i \in \mathcal{Z}} \sum_{j \in \mathcal{Z}} x_{ij} \le \sum_{j \in \mathcal{Z}} d_j$. Thus, $\mathcal{X}(\boldsymbol{b})$ is nonempty and compact and lies within a common compact set. Because $\boldsymbol{s}(\boldsymbol{b})$ is continuous and every defining inequality is continuous in $(\boldsymbol{b}, \boldsymbol{x})$, $\mathcal{X}(\cdot)$ has a closed graph and is upper hemicontinuous.

It remains to establish lower hemicontinuity. Let $\boldsymbol{b}^\nu \in \mathcal{B}$ converge to $\boldsymbol{b}$, and take any $\boldsymbol{x} \in \mathcal{X}(\boldsymbol{b})$. For each i, let a_i^ν be the fraction of its matching flows retained to satisfy the supply limit at $\boldsymbol{b}^\nu$. Set $a_i^\nu := \min\{1, s_i(\boldsymbol{b}^\nu)/q_i(\boldsymbol{x})\}$ if $q_i(\boldsymbol{x}) > 0$ and $a_i^\nu := 1$ otherwise, and define $\widetilde{x}_{ij}^\nu := a_i^\nu x_{ij}$. If $q_i(\boldsymbol{x}) > 0$, feasibility implies $s_i(\boldsymbol{b}) \ge q_i(\boldsymbol{x}) > 0$, and the continuity of s_i gives $a_i^\nu \to 1$. The same conclusion follows immediately when $q_i(\boldsymbol{x}) = 0$. Hence, $\widetilde{\boldsymbol{x}}^\nu \to \boldsymbol{x}$. Row scaling preserves the demand constraints and ensures $q_i(\widetilde{\boldsymbol{x}}^\nu) \le s_i(\boldsymbol{b}^\nu)$.

To restore the parameter-dependent budget constraint, adopt the convention $B/0 = +\infty$ and define $\theta_\nu := \min\left\{1, \frac{B}{(\boldsymbol{b}^\nu)^\top \boldsymbol{q}(\widetilde{\boldsymbol{x}}^\nu)}\right\}$, $\boldsymbol{x}^\nu := \theta_\nu \widetilde{\boldsymbol{x}}^\nu$. The denominator converges to $\boldsymbol{b}^\top \boldsymbol{q}(\boldsymbol{x}) \le B$. Since $B > 0$, it follows that $\theta_\nu \to 1$. Therefore, $\boldsymbol{x}^\nu \in \mathcal{X}(\boldsymbol{b}^\nu)$ and $\boldsymbol{x}^\nu \to \boldsymbol{x}$, which proves lower hemicontinuity. The matching objective is continuous in $(\boldsymbol{b}, \boldsymbol{x})$. Berge's maximum theorem (Berge 1963) therefore implies that $Z(\boldsymbol{b})$ is continuous on $\mathcal{B}$. Since $\mathcal{B}$ is compact, existence follows from the Weierstrass theorem (Rudin 1976). □

EC.4.2. Proof of Proposition 3

Proof. Define the convex hull

$$\mathcal{H}_k := \operatorname{conv}\left\{\nabla Z(\boldsymbol{y}^{k,a}) : a \in \mathcal{A}_k\right\}. \tag{EC.11}$$

Every ambient gradient in Eq. (EC.11) belongs to $\partial^C Z(\boldsymbol{y}^{k,a})$, and every ambient point lies within distance ϵ_k of $\boldsymbol{b}^k$. It follows from Eq. (18) that $\mathcal{H}_k \subseteq \partial^C_{\epsilon_k} Z(\boldsymbol{b}^k)$. Because the columns of $\boldsymbol{N}_k$ are exactly the active outward normals, up to duplicate generators, $\{\boldsymbol{N}_k \boldsymbol{\mu} : \boldsymbol{\mu} \ge \boldsymbol{0}\} = \mathcal{N}_{\mathcal{B}}(\boldsymbol{b}^k)$. Consequently, Eq. (19) equals $\operatorname{dist}(\mathcal{H}_k, \mathcal{N}_{\mathcal{B}}(\boldsymbol{b}^k))$. Expanding the first set from $\mathcal{H}_k$ to $\partial^C_{\epsilon_k} Z(\boldsymbol{b}^k)$ can only decrease the set distance, which proves Eq. (20). □

EC.4.3. Sampled Feasible Ascent Direction

The finite-scale certificate uses only ambient gradients. The next lemma returns to the working bundle and establishes the geometry of its search direction.

LEMMA EC.1 **(Sampled feasible ascent direction).** *Let $\boldsymbol{b}^k \in \mathcal{B}$, and let $(\boldsymbol{\omega}^k, \boldsymbol{\mu}^k)$ solve Eq. (EC.9). Then the residual direction defined in Eq. (EC.10) satisfies*

$$\boldsymbol{N}_k^\top \boldsymbol{r}^k \le \boldsymbol{0}, \quad (\boldsymbol{\mu}^k)^\top \boldsymbol{N}_k^\top \boldsymbol{r}^k = 0, \tag{EC.12}$$

$$\min_{a \in \mathcal{I}_k} (\boldsymbol{g}^{k,a})^\top \boldsymbol{r}^k \ge \|\boldsymbol{r}^k\|_2^2. \tag{EC.13}$$

Consequently, $\boldsymbol{r}^k$ belongs to the tangent cone $T_{\mathcal{B}}(\boldsymbol{b}^k)$. If $\boldsymbol{r}^k \ne \boldsymbol{0}$, every sensitivity in the working bundle has strictly positive first-order gain along $\boldsymbol{r}^k$.

Proof. The optimality conditions with respect to $\boldsymbol{\mu}$ give $\boldsymbol{N}_k^\top \boldsymbol{r}^k \le \boldsymbol{0}$, and complementary slackness gives the second relation in Eq. (EC.12). The simplex optimality conditions for $\boldsymbol{\omega}$ give $(\boldsymbol{g}^{k,a})^\top \boldsymbol{r}^k \ge (\bar{\boldsymbol{g}}^k)^\top \boldsymbol{r}^k$ for all $a \in \mathcal{I}_k$. Moreover, $(\bar{\boldsymbol{g}}^k)^\top \boldsymbol{r}^k = (\boldsymbol{r}^k + \bar{\boldsymbol{n}}^k)^\top \boldsymbol{r}^k = \|\boldsymbol{r}^k\|_2^2$, because $(\bar{\boldsymbol{n}}^k)^\top \boldsymbol{r}^k = (\boldsymbol{\mu}^k)^\top \boldsymbol{N}_k^\top \boldsymbol{r}^k = 0$. This proves Eq. (EC.13). For the polyhedral set $\mathcal{B}$, the tangent cone at $\boldsymbol{b}^k$ is $T_{\mathcal{B}}(\boldsymbol{b}^k) = \{\boldsymbol{v} : \boldsymbol{N}_k^\top \boldsymbol{v} \le \boldsymbol{0}\}$. □

EC.4.4. Convergence of the Smooth Specialization

We analyze the center-gradient specialization defined in Section EC.3.4 of the e-companion. Assume the conditions of Lemma 1 hold, and let

$$\mathcal{L}_0 := \{\boldsymbol{b} \in \mathcal{B} : Z(\boldsymbol{b}) \ge Z(\boldsymbol{b}^0)\}.$$

Throughout this subsection, assume $B > 0$ and that the stable optimal-basis condition in Section 4.1 holds on an open neighborhood of $\mathcal{L}_0$. Let $\bar{\tau} > 0$, $\delta \in (0,1)$, and $\chi \in (0,1)$ denote the initial trial step, backtracking contraction factor, and Armijo parameter, respectively.

A point $\bar{\boldsymbol{b}} \in \mathcal{B}$ is stationary if

$$\nabla Z(\bar{\boldsymbol{b}})^\top (\boldsymbol{b} - \bar{\boldsymbol{b}}) \le 0, \qquad \forall \boldsymbol{b} \in \mathcal{B}.$$

For a fixed reference step $\tau_{\text{ref}} > 0$, define the projected-gradient residual

$$R_{\tau_{\text{ref}}}(\boldsymbol{b}) := \frac{1}{\tau_{\text{ref}}} \left\| \Pi_{\mathcal{B}}\big(\boldsymbol{b} + \tau_{\text{ref}} \nabla Z(\boldsymbol{b})\big) - \boldsymbol{b} \right\|_2. \tag{EC.14}$$

PROPOSITION EC.1 **(Convergence of the smooth specialization)**. *The projected Armijo procedure in Section EC.3.4 of the e-companion terminates finitely at each iteration. For the resulting infinite sequence,*

$$\lim_{k \to \infty} R_{\tau_{\text{ref}}}(\boldsymbol{b}^k) = 0 \tag{EC.15}$$

for every fixed $\tau_{\text{ref}} > 0$, and every accumulation point is stationary.

Proof. The stable optimal-basis condition and standard parametric-LP sensitivity analysis (Gal and Greenberg 1997) imply that Z is continuously differentiable with a Lipschitz-continuous gradient on a closed neighborhood of $\mathcal{L}_0$. The envelope formula identifies the center sensitivity with $\nabla Z(\boldsymbol{b})$.

For a trial step $\tau > 0$, define $\boldsymbol{g}^k := \nabla Z(\boldsymbol{b}^k), \boldsymbol{h}^k(\tau) := \Pi_{\mathcal{B}}(\boldsymbol{b}^k + \tau \boldsymbol{g}^k) - \boldsymbol{b}^k$. Projection optimality then gives

$$(\boldsymbol{g}^k)^\top \boldsymbol{h}^k(\tau) \ge \frac{1}{\tau} \|\boldsymbol{h}^k(\tau)\|_2^2. \tag{EC.16}$$

Since $\mathcal{L}_0$ is compact and contained in the open stable-basis neighborhood, choose $r > 0$ such that its closed r-neighborhood remains within that neighborhood. Let L be a Lipschitz constant of ∇Z on this closed neighborhood, and let $M := \max_{\boldsymbol{b} \in \mathcal{L}_0} \|\nabla Z(\boldsymbol{b})\|_2$. For $\boldsymbol{b}^k \in \mathcal{L}_0$, the nonexpansiveness of the projection gives $\|\boldsymbol{h}^k(\tau)\|_2 \le \tau \|\boldsymbol{g}^k\|_2 \le \tau M$. The ascent lemma and Eq. (EC.16) imply

$$\begin{aligned} Z(\boldsymbol{b}^k + \boldsymbol{h}^k(\tau)) &\ge Z(\boldsymbol{b}^k) + (\boldsymbol{g}^k)^\top \boldsymbol{h}^k(\tau) - \frac{L}{2} \|\boldsymbol{h}^k(\tau)\|_2^2 \\ &\ge Z(\boldsymbol{b}^k) + \chi (\boldsymbol{g}^k)^\top \boldsymbol{h}^k(\tau) \end{aligned}$$

whenever $\tau \le \widehat{\tau} := \min\left\{ \frac{2(1-\chi)}{L}, \frac{r}{M} \right\}$. Here, $r/M := +\infty$ if $M = 0$, and the first ratio is interpreted as $+\infty$ if $L = 0$. Let τ_k denote the first accepted trial step. Thus, backtracking terminates finitely, and every accepted step satisfies

$$\tau_k \ge \underline{\tau} := \min\{\bar{\tau}, \delta \widehat{\tau}\} > 0. \tag{EC.17}$$

For the accepted step, set $\boldsymbol{h}^k := \boldsymbol{h}^k(\tau_k) = \boldsymbol{b}^{k+1} - \boldsymbol{b}^k$. The Armijo condition, Eq. (EC.16), and $\tau_k \le \bar{\tau}$ yield

$$Z(\boldsymbol{b}^{k+1}) - Z(\boldsymbol{b}^k) \ge \chi (\boldsymbol{g}^k)^\top \boldsymbol{h}^k \ge \frac{\chi}{\bar{\tau}} \|\boldsymbol{h}^k\|_2^2. \tag{EC.18}$$

Since $\boldsymbol{b}^0 \in \mathcal{L}_0$, the accepted iterates remain in $\mathcal{L}_0$ by induction. By Lemma 1, Z is bounded above on the compact set $\mathcal{B}$. Summing Eq. (EC.18) gives $\sum_{k=0}^{\infty} \|\boldsymbol{h}^k\|_2^2 < \infty$, and therefore $\boldsymbol{h}^k \to \mathbf{0}$.

Let $\boldsymbol{b}^{k_\nu} \to \bar{\boldsymbol{b}}$ be any convergent subsequence. By Eq. (EC.17), a further subsequence satisfies $\tau_{k_\nu} \to \bar{\tau}_0 \in [\underline{\tau}, \bar{\tau}]$. The continuity of the gradient and projection, together with $\boldsymbol{b}^{k_\nu+1} = \Pi_{\mathcal{B}}\big(\boldsymbol{b}^{k_\nu} + \tau_{k_\nu} \nabla Z(\boldsymbol{b}^{k_\nu})\big)$ and $\boldsymbol{h}^{k_\nu} \to \mathbf{0}$, gives $\bar{\boldsymbol{b}} = \Pi_{\mathcal{B}}\big(\bar{\boldsymbol{b}} + \bar{\tau}_0 \nabla Z(\bar{\boldsymbol{b}})\big)$. Since $\bar{\tau}_0 > 0$, the projection variational inequality yields $\nabla Z(\bar{\boldsymbol{b}})^\top (\boldsymbol{b} - \bar{\boldsymbol{b}}) \le 0, \forall \boldsymbol{b} \in \mathcal{B}$. Thus, every accumulation point is stationary.

For any fixed $\tau_{\mathrm{ref}} > 0$, the residual $R_{\tau_{\mathrm{ref}}}$ is continuous on $\mathcal{L}_0$ and vanishes exactly at stationary points. Suppose, to the contrary, that Eq. (EC.15) fails. Compactness then yields a subsequence with residuals bounded away from zero and a stationary accumulation point, contradicting the continuity of $R_{\tau_{\mathrm{ref}}}$. Therefore, Eq. (EC.15) holds for every fixed $\tau_{\mathrm{ref}} > 0$. □

Corollary EC.1 **(Complexity of accepted smooth steps)**. *Under the assumptions of Proposition EC.1, define the projected-gradient mapping*

$$\mathcal{G}_{\tau_k}(\boldsymbol{b}^k) := \frac{\Pi_{\mathcal{B}}\big(\boldsymbol{b}^k + \tau_k \nabla Z(\boldsymbol{b}^k)\big) - \boldsymbol{b}^k}{\tau_k}. \tag{EC.19}$$

For every integer $K \ge 1$,

$$\min_{0 \le k < K} \big\|\mathcal{G}_{\tau_k}(\boldsymbol{b}^k)\big\|_2^2 \le \frac{Z^\star - Z(\boldsymbol{b}^0)}{\chi \underline{\tau} K}, \tag{EC.20}$$

where $\underline{\tau} > 0$ is defined in Eq. (EC.17). Thus, the smooth specialization reaches an iterate with gradient-mapping norm at most ε within $O(\varepsilon^{-2})$ accepted steps.

Proof. For an accepted step, the Armijo condition and Eqs. (EC.16) and (EC.17) give

$$Z(\boldsymbol{b}^{k+1}) - Z(\boldsymbol{b}^k) \ge \chi \tau_k \big\|\mathcal{G}_{\tau_k}(\boldsymbol{b}^k)\big\|_2^2 \ge \chi \underline{\tau} \big\|\mathcal{G}_{\tau_k}(\boldsymbol{b}^k)\big\|_2^2. \tag{EC.21}$$

Summing Eq. (EC.21) for $k = 0, \dots, K-1$ and using $Z(\boldsymbol{b}^K) \le Z^\star$ proves Eq. (EC.20). □

EC.4.5. Proof of Proposition 4

Proof. For each ℓ, choose minimizers $(\boldsymbol{\omega}^\ell, \boldsymbol{\mu}^\ell)$ defining $\widehat{R}_\ell^{\mathrm{amb}}$, and set $\bar{\boldsymbol{g}}^\ell := \boldsymbol{G}_\ell^{\mathrm{amb}} \boldsymbol{\omega}^\ell$ and $\bar{\boldsymbol{n}}^\ell := \boldsymbol{N}_\ell \boldsymbol{\mu}^\ell$. Since $\boldsymbol{b}^\ell \to \bar{\boldsymbol{b}}$ and $\epsilon_\ell \to 0$, local Lipschitz continuity bounds the sampled gradients for all sufficiently large ℓ, so $\{\bar{\boldsymbol{g}}^\ell\}$ is bounded. Moreover, $\|\bar{\boldsymbol{g}}^\ell - \bar{\boldsymbol{n}}^\ell\|_2 = \widehat{R}_\ell^{\mathrm{amb}} \to 0$, so $\{\bar{\boldsymbol{n}}^\ell\}$ is also bounded. Passing to a subsequence, we obtain $\bar{\boldsymbol{g}}^\ell \to \bar{\boldsymbol{g}}$, $\bar{\boldsymbol{n}}^\ell \to \bar{\boldsymbol{n}}$, and $\bar{\boldsymbol{g}} = \bar{\boldsymbol{n}}$.

By Carathéodory's theorem, each $\bar{\boldsymbol{g}}^\ell$ can be represented using at most $|\mathcal{Z}| + 1$ columns of $\boldsymbol{G}_\ell^{\mathrm{amb}}$. The corresponding sample points satisfy $\|\boldsymbol{y}^{\ell,a} - \bar{\boldsymbol{b}}\|_2 \le \epsilon_\ell + \|\boldsymbol{b}^\ell - \bar{\boldsymbol{b}}\|_2 \to 0$. Boundedness of the gradients and compactness of the coefficient simplex yield a subsequence along which all these terms converge. The gradient-limit characterization and convexity of the Clarke generalized gradient therefore give $\bar{\boldsymbol{g}} \in \partial^C Z(\bar{\boldsymbol{b}})$.

For every $\widetilde{\boldsymbol{b}} \in \mathcal{B}$, the representation in terms of active outward normals gives $(\bar{\boldsymbol{n}}^\ell)^\top (\widetilde{\boldsymbol{b}} - \boldsymbol{b}^\ell) \le 0$. Taking limits yields $\bar{\boldsymbol{n}}^\top (\widetilde{\boldsymbol{b}} - \bar{\boldsymbol{b}}) \le 0$ for every $\widetilde{\boldsymbol{b}} \in \mathcal{B}$, and hence $\bar{\boldsymbol{n}} \in \mathcal{N}_{\mathcal{B}}(\bar{\boldsymbol{b}})$. Since $\bar{\boldsymbol{g}} = \bar{\boldsymbol{n}}$, we obtain $\partial^C Z(\bar{\boldsymbol{b}}) \cap \mathcal{N}_{\mathcal{B}}(\bar{\boldsymbol{b}}) \ne \varnothing$, equivalently, $\mathbf{0} \in -\partial^C Z(\bar{\boldsymbol{b}}) + \mathcal{N}_{\mathcal{B}}(\bar{\boldsymbol{b}})$. Thus, $\bar{\boldsymbol{b}}$ is a constrained Clarke-stationary point of DSMDP. □

REMARK EC.1 (FINITE AND LIMITING ROLES OF THE AMBIENT RESIDUAL). For a fixed sampling radius, Proposition 3 uses the ambient residual $\widehat{R}_k^{\mathrm{amb}}$ to upper-bound the enlarged-Clarke stationarity distance. The certificate applies at a fixed sampling scale and is therefore reported together with its radius. Proposition 4 yields exact Clarke stationarity only along convergent ambient assessments whose radii and residuals both vanish.

EC.5. Conic DSMDP Relaxation

Using valid bounds on the MNL utilities and denominators, we construct a conic outer relaxation based on exponential-cone epigraphs and affine secant upper bounds for the response terms and McCormick envelopes for the remaining products. The complete formulation contains a lift of every feasible DSMDP solution and distinguishes the solver-certified upper bound from the feasible Conic benchmark used in the experiments.

EC.5.1. Bounds for the MNL Utilities and Denominators

Each normalized MNL utility depends affinely on a single bonus coordinate, so valid bounds over $\mathcal{B}$ follow by evaluating the utility at 0 and $b^{\max}$. For $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$, this gives $\underline{V}_{ij} = \beta^{(3)} d_{ij} + \min\{0, \beta^{(1)} b^{\max}\}$ and $\overline{V}_{ij} = \beta^{(3)} d_{ij} + \max\{0, \beta^{(1)} b^{\max}\}$, while $\underline{V}_i^{\mathrm{Off}} = \delta_i^{\mathrm{Off}} + \min\{0, -\zeta^{\mathrm{Off}} b^{\max}\}$ and $\overline{V}_i^{\mathrm{Off}} = \delta_i^{\mathrm{Off}} + \max\{0, -\zeta^{\mathrm{Off}} b^{\max}\}$ for $i \in \mathcal{Z}$. These bounds imply $\underline{t}_{ij} = \exp(\underline{V}_{ij})$ and $\overline{t}_{ij} = \exp(\overline{V}_{ij})$ for $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$, and $\underline{t}_i^{\mathrm{Off}} = \exp(\underline{V}_i^{\mathrm{Off}})$ and $\overline{t}_i^{\mathrm{Off}} = \exp(\overline{V}_i^{\mathrm{Off}})$ for $i \in \mathcal{Z}$. The denominator variable satisfies $D_i \in [\underline{D}_i, \overline{D}_i]$, where $\underline{D}_i = \sum_{m \in \mathcal{Z}_i} \underline{t}_{im} + \underline{t}_i^{\mathrm{Off}}$ and $\overline{D}_i = \sum_{m \in \mathcal{Z}_i} \overline{t}_{im} + \overline{t}_i^{\mathrm{Off}}$ for all $i \in \mathcal{Z}$.

For any finite interval $[\ell, u]$, the affine secant introduced in Section 5.3 is

$$\mathcal{S}_{\exp}(v; \ell, u) := \begin{cases} \dfrac{u - v}{u - \ell} \exp(\ell) + \dfrac{v - \ell}{u - \ell} \exp(u), & \ell < u, \\ \exp(\ell), & \ell = u. \end{cases} \tag{EC.22}$$

By convexity, $\exp(v) \le \mathcal{S}_{\exp}(v; \ell, u)$ for all $v \in [\ell, u]$. When $\ell = u$, the exponential lower bound and secant upper bound coincide.

EC.5.2. Conic Relaxation of the MNL Probabilities

Using the variables introduced in Section 5.3, impose $D_i = \sum_{m \in \mathcal{Z}_i} t_{im} + t_i^{\mathrm{Off}} \in [\underline{D}_i, \overline{D}_i]$ for each $i \in \mathcal{Z}$.

For each $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$, the exponential and box constraints are

$$t_{ij} \ge \exp(V_{ij}), \tag{EC.23}$$

$$t_{ij} \le \mathcal{S}_{\exp}\big(V_{ij}; \underline{V}_{ij}, \overline{V}_{ij}\big), \tag{EC.24}$$

$$\underline{t}_{ij} \le t_{ij} \le \overline{t}_{ij}. \tag{EC.25}$$

The probability variables satisfy

$$\begin{aligned} \sum_{j \in \mathcal{Z}_i} P_{ij} + P_i^{\mathrm{Off}} = 1, \quad & i \in \mathcal{Z}, \\ P_{ij} \ge 0, \quad & i \in \mathcal{Z}, \quad j \in \mathcal{Z}_i, \\ P_i^{\mathrm{Off}} \ge 0, \quad & i \in \mathcal{Z}. \end{aligned} \tag{EC.26}$$

The exact probability–denominator relations are

$$\begin{aligned} t_{ij} &= P_{ij} D_i, \qquad i \in \mathcal{Z},\ j \in \mathcal{Z}_i, \\ t_i^{\mathrm{Off}} &= P_i^{\mathrm{Off}} D_i, \qquad i \in \mathcal{Z}. \end{aligned} \tag{EC.27}$$

Equation (EC.26) implies $P_{ij} \in [0,1]$ and $P_i^{\mathrm{Off}} \in [0,1]$. Together with $D_i \in [\underline{D}_i, \overline{D}_i]$, these bounds yield the following McCormick envelopes.

For each $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$, the envelope for $t_{ij} = P_{ij} D_i$ is

$$t_{ij} \geq \underline{D}_i P_{ij}, \tag{EC.28}$$

$$t_{ij} \geq D_i + \overline{D}_i (P_{ij} - 1), \tag{EC.29}$$

$$t_{ij} \leq \overline{D}_i P_{ij}, \tag{EC.30}$$

$$t_{ij} \leq D_i + \underline{D}_i (P_{ij} - 1). \tag{EC.31}$$

For each $i \in \mathcal{Z}$, the envelope for $t_i^{\mathrm{Off}} = P_i^{\mathrm{Off}} D_i$ is

$$t_i^{\mathrm{Off}} \geq \underline{D}_i P_i^{\mathrm{Off}}, \tag{EC.32}$$

$$t_i^{\mathrm{Off}} \geq D_i + \overline{D}_i (P_i^{\mathrm{Off}} - 1), \tag{EC.33}$$

$$t_i^{\mathrm{Off}} \leq \overline{D}_i P_i^{\mathrm{Off}}, \tag{EC.34}$$

$$t_i^{\mathrm{Off}} \leq D_i + \underline{D}_i (P_i^{\mathrm{Off}} - 1). \tag{EC.35}$$

Equations (EC.28)–(EC.35) define the convex hulls of the graphs $t_{ij} = P_{ij} D_i$ and $t_i^{\mathrm{Off}} = P_i^{\mathrm{Off}} D_i$ over their respective boxes.

EC.5.3. Conic Relaxation of Latent Driver Activation and the Unavailable State Exponential Term

We express the latent-driver activation probability using a binary logit model. Define $U_i^{\mathrm{On}} := \alpha_i^{\mathrm{On}} + \gamma^{\mathrm{On}} b_i$ and the negative utility $\widetilde{U}_i^{\mathrm{On}} := -U_i^{\mathrm{On}}$. Introduce the exponential variable e_i^{On}, the denominator variable D_i^{On}, and the probability variable φ_i^{On} representing $P_i^{\mathrm{On}}(\boldsymbol{b})$. Their exact relations are $e_i^{\mathrm{On}} = \exp(\widetilde{U}_i^{\mathrm{On}})$,, $D_i^{\mathrm{On}} = 1 + e_i^{\mathrm{On}}$,, and $1 = \varphi_i^{\mathrm{On}} D_i^{\mathrm{On}}$. The endpoint utility bounds are $\underline{U}_i^{\mathrm{On}} = \min\{\alpha_i^{\mathrm{On}}, \alpha_i^{\mathrm{On}} + \gamma^{\mathrm{On}} b^{\max}\}$, and $\overline{U}_i^{\mathrm{On}} = \max\{\alpha_i^{\mathrm{On}}, \alpha_i^{\mathrm{On}} + \gamma^{\mathrm{On}} b^{\max}\}, \forall i \in \mathcal{Z}$. For $\gamma^{\mathrm{On}} \geq 0$, these bounds reduce to $\underline{U}_i^{\mathrm{On}} = \alpha_i^{\mathrm{On}}, \quad \overline{U}_i^{\mathrm{On}} = \alpha_i^{\mathrm{On}} + \gamma^{\mathrm{On}} b^{\max}$. The negative-utility bounds are $\underline{\widetilde{U}}_i^{\mathrm{On}} = -\overline{U}_i^{\mathrm{On}}$, and $\overline{\widetilde{U}}_i^{\mathrm{On}} = -\underline{U}_i^{\mathrm{On}}$. The corresponding exponential, denominator, and probability bounds are $\underline{e}_i^{\mathrm{On}} := \exp(\underline{\widetilde{U}}_i^{\mathrm{On}}) = \exp(-\overline{U}_i^{\mathrm{On}})$ and $\overline{e}_i^{\mathrm{On}} := \exp(\overline{\widetilde{U}}_i^{\mathrm{On}}) = \exp(-\underline{U}_i^{\mathrm{On}})$; $\underline{D}_i^{\mathrm{On}} := 1 + \underline{e}_i^{\mathrm{On}}$ and $\overline{D}_i^{\mathrm{On}} := 1 + \overline{e}_i^{\mathrm{On}}$; and $\underline{\varphi}_i^{\mathrm{On}} := \sigma(\underline{U}_i^{\mathrm{On}})$ and $\overline{\varphi}_i^{\mathrm{On}} := \sigma(\overline{U}_i^{\mathrm{On}})$, for all $i \in \mathcal{Z}$.

For each $i \in \mathcal{Z}$, we use an exponential-cone epigraph, an affine secant upper bound, and a McCormick envelope. The epigraph and secant constraints enclose the exponential relation over the valid utility interval. The McCormick inequalities relax the reciprocal product between the activation probability and its denominator.

$$e_i^{\mathrm{On}} \geq \exp(\widetilde{U}_i^{\mathrm{On}}), \tag{EC.36}$$

$$e_i^{\mathrm{On}} \leq \mathcal{S}_{\exp}\big(\widetilde{U}_i^{\mathrm{On}}; \underline{\widetilde{U}}_i^{\mathrm{On}}, \overline{\widetilde{U}}_i^{\mathrm{On}}\big), \tag{EC.37}$$

$$D_i^{\mathrm{On}} = 1 + e_i^{\mathrm{On}}, \tag{EC.38}$$
$$\underline{D}_i^{\mathrm{On}} \le D_i^{\mathrm{On}} \le \overline{D}_i^{\mathrm{On}}, \tag{EC.39}$$
$$\underline{\varphi}_i^{\mathrm{On}} \le \varphi_i^{\mathrm{On}} \le \overline{\varphi}_i^{\mathrm{On}}, \tag{EC.40}$$
$$1 \ge \underline{\varphi}_i^{\mathrm{On}} D_i^{\mathrm{On}} + \underline{D}_i^{\mathrm{On}} \varphi_i^{\mathrm{On}} - \underline{\varphi}_i^{\mathrm{On}} \underline{D}_i^{\mathrm{On}}, \tag{EC.41}$$
$$1 \ge \overline{\varphi}_i^{\mathrm{On}} D_i^{\mathrm{On}} + \overline{D}_i^{\mathrm{On}} \varphi_i^{\mathrm{On}} - \overline{\varphi}_i^{\mathrm{On}} \overline{D}_i^{\mathrm{On}}, \tag{EC.42}$$
$$1 \le \overline{\varphi}_i^{\mathrm{On}} D_i^{\mathrm{On}} + \underline{D}_i^{\mathrm{On}} \varphi_i^{\mathrm{On}} - \overline{\varphi}_i^{\mathrm{On}} \underline{D}_i^{\mathrm{On}}, \tag{EC.43}$$
$$1 \le \underline{\varphi}_i^{\mathrm{On}} D_i^{\mathrm{On}} + \overline{D}_i^{\mathrm{On}} \varphi_i^{\mathrm{On}} - \underline{\varphi}_i^{\mathrm{On}} \overline{D}_i^{\mathrm{On}}. \tag{EC.44}$$

Every exact binary logit point satisfies Eqs. (EC.36)–(EC.44). These constraints therefore define an outer relaxation of the latent-driver activation response. For each $i \in \mathcal{Z}$, the exponential term t_i^{Off} satisfies the following bounds.

$$t_i^{\mathrm{Off}} \ge \exp(V_i^{\mathrm{Off}}), \tag{EC.45}$$
$$t_i^{\mathrm{Off}} \le \mathcal{S}_{\exp}\big(V_i^{\mathrm{Off}}; \underline{V}_i^{\mathrm{Off}}, \overline{V}_i^{\mathrm{Off}}\big), \tag{EC.46}$$
$$\underline{t}_i^{\mathrm{Off}} \le t_i^{\mathrm{Off}} \le \overline{t}_i^{\mathrm{Off}}. \tag{EC.47}$$

Together with the McCormick relaxation of $t_i^{\mathrm{Off}} = P_i^{\mathrm{Off}} D_i$, these constraints retain every exact departure from the dispatchable pool and latent-driver activation response within the lifted feasible region.

EC.5.4. McCormick Relaxation of Bonus Payments

For the matched quantity q_i defined in Section 5.3, combining total demand with the maximum possible post-response supply gives the upper bound

$$\bar{q}_i = \min\left\{ \sum_{j \in \mathcal{Z}} d_j,\ s_i^{\mathrm{On}} + \sum_{j \in \mathcal{Z}_i \setminus \{i\}} s_j^{\mathrm{On}} + s_i^{\mathrm{Off}} \right\}, \qquad \forall i \in \mathcal{Z}, \tag{EC.48}$$
$$0 \le q_i \le \bar{q}_i, \qquad \forall i \in \mathcal{Z}. \tag{EC.49}$$

Under the symmetric reachability relation and $i \in \mathcal{Z}_i$, the dispatchable-supply term in Eq. (EC.48) can equivalently be written as $\sum_{j \in \mathcal{Z}:\, i \in \mathcal{Z}_j} s_j^{\mathrm{On}}$. For the New York City instances, every sum involving x_{ij} is restricted to the sparse matching arc set described in Section 6.8; the original model and its outer relaxation use the same arc set.

Over $(b_i, q_i) \in [0, b^{\max}] \times [0, \bar{q}_i]$, the McCormick envelope for $w_i = b_i q_i$ is

$$w_i \ge 0, \tag{EC.50}$$
$$w_i \ge b^{\max} q_i + \bar{q}_i b_i - b^{\max} \bar{q}_i, \tag{EC.51}$$
$$w_i \le b^{\max} q_i, \tag{EC.52}$$
$$w_i \le \bar{q}_i b_i, \qquad \forall i \in \mathcal{Z}. \tag{EC.53}$$

Equations (EC.50)–(EC.53) give the McCormick envelope of the payment product.

EC.5.5. Full Conic Relaxation

Collect the decision variables in $\boldsymbol{\vartheta} := (\boldsymbol{x}, \boldsymbol{q}, \boldsymbol{w}, \boldsymbol{b}, \boldsymbol{P}, \boldsymbol{P}^{\mathrm{Off}}, \boldsymbol{t}, \boldsymbol{t}^{\mathrm{Off}}, (D_i)_{i\in\mathcal{Z}}, \boldsymbol{\varphi}^{\mathrm{On}}, \boldsymbol{e}^{\mathrm{On}}, \boldsymbol{D}^{\mathrm{On}})$. The full conic relaxation is

$$
\begin{aligned}
(\overline{\mathrm{DSMDP}}) \qquad \overline{Z} = \max_{\boldsymbol{\vartheta}} \quad & \sum_{i,j\in\mathcal{Z}} p_{ij}x_{ij} - \sum_{i\in\mathcal{Z}} w_i && && \text{(EC.54a)}\\
\text{s.t.} \quad & \sum_{i\in\mathcal{Z}} x_{ij} \le d_j, && \forall j\in\mathcal{Z}, && \text{(EC.54b)}\\
& \sum_{j\in\mathcal{Z}} x_{ij} \le s_i^{\mathrm{On}} + \sum_{j\in\mathcal{Z}_i\setminus\{i\}} s_j^{\mathrm{On}} P_{ji} \\
& \qquad - s_i^{\mathrm{On}} \left(\sum_{j\in\mathcal{Z}_i\setminus\{i\}} P_{ij} + P_i^{\mathrm{Off}} \right) + s_i^{\mathrm{Off}} \varphi_i^{\mathrm{On}}, && \forall i\in\mathcal{Z}. && \text{(EC.54c)}\\
& \sum_{i\in\mathcal{Z}} w_i \le B, && && \text{(EC.54d)}\\
& q_i = \sum_{j\in\mathcal{Z}} x_{ij}, && i\in\mathcal{Z}, && \text{(EC.54e)}\\
& \text{(EC.49)}, \qquad \text{(EC.50)} - - \text{(EC.53)}, && && \text{(EC.54f)}\\
& \sum_{j\in\mathcal{Z}_i} P_{ij} + P_i^{\mathrm{Off}} = 1, && i\in\mathcal{Z}, && \text{(EC.54g)}\\
& P_{ij} \ge 0, && i\in\mathcal{Z}, \quad j\in\mathcal{Z}_i, && \text{(EC.54h)}\\
& P_i^{\mathrm{Off}} \ge 0, && i\in\mathcal{Z}, && \text{(EC.54i)}\\
& D_i = \sum_{m\in\mathcal{Z}_i} t_{im} + t_i^{\mathrm{Off}}, && i\in\mathcal{Z}, && \text{(EC.54j)}\\
& \underline{D}_i \le D_i \le \overline{D}_i, && i\in\mathcal{Z}, && \text{(EC.54k)}\\
& \text{(EC.23)} - - \text{(EC.25)}, \qquad \text{(EC.45)} - - \text{(EC.47)}, && && \text{(EC.54l)}\\
& \text{(EC.28)} - - \text{(EC.35)}, && && \text{(EC.54m)}\\
& \text{(EC.36)} - - \text{(EC.44)}, && && \text{(EC.54n)}\\
& \boldsymbol{b}\in\mathcal{B}, && && \text{(EC.54o)}\\
& x_{ij} \ge 0, && i,j\in\mathcal{Z}. && \text{(EC.54p)}
\end{aligned}
$$

Objective (EC.54a) maximizes the relaxed net operating profit. Constraints (EC.54b) and (EC.54c) impose demand capacity and the relaxed post-response supply balance. Constraint (EC.54d) enforces the bonus budget, Constraint (EC.54e) defines matched quantity, and Constraint (EC.54f) invokes the bounds on matched quantities and payment envelopes. Constraint (EC.54g) normalizes the relocation and unavailability probabilities, while Constraints (EC.54h) and (EC.54i) impose their domains. Constraints (EC.54j) and (EC.54k) define and bound the MNL denominators. Constraints (EC.54l)–(EC.54m) impose the exponential and product relaxations for relocation and unavailability, whereas Constraint (EC.54n) imposes the corresponding relaxation for latent-driver activation. Constraints (EC.54o) and (EC.54p) retain the feasible driver surge maps and nonnegative matching flows.

EC.5.6. Validity of the Upper Bound

The upper-bound proof constructs a feasible lift of an arbitrary DSMDP solution into $\overline{\text{DSMDP}}$ without changing its objective value. The feasible-set inclusion establishes the bound independently of the feasible-recovery procedure used for the Conic benchmark.

PROPOSITION EC.2 **(Validity of the conic upper bound).** *Let $Z^\star$ denote the optimal objective value of the original DSMDP, and let $\overline{Z}$ denote the optimal objective value of the conic outer relaxation $\overline{\text{DSMDP}}$ in Eq. (EC.54). Then $Z^\star \le \overline{Z}$.*

Proof. Consider any feasible solution $(\boldsymbol{b}, \boldsymbol{x})$ of the original DSMDP. For each $i \in \mathcal{Z}$ and $j \in \mathcal{Z}_i$, set $t_{ij} = \exp(V_{ij})$, $t_i^{\text{Off}} = \exp(V_i^{\text{Off}})$, and $D_i = \sum_{m \in \mathcal{Z}_i} t_{im} + t_i^{\text{Off}}$. Take $P_{ij} = P_{ij}(\boldsymbol{b}) = t_{ij}/D_i$ and $P_i^{\text{Off}} = P_i^{\text{Off}}(\boldsymbol{b}) = t_i^{\text{Off}}/D_i$. These assignments satisfy the probability normalization, denominator definition, exponential-cone epigraph constraints, and all prescribed bounds. By convexity of the exponential function, the exact exponential values also satisfy the secant upper bounds in Eqs. (EC.24) and (EC.46). The assignments further satisfy $t_{ij} = P_{ij} D_i$ and $t_i^{\text{Off}} = P_i^{\text{Off}} D_i$. Every point on either bilinear graph satisfies its McCormick envelope, so all MNL outer-relaxation constraints hold. For the latent-driver activation response, set $U_i^{\text{On}} = \alpha_i^{\text{On}} + \gamma^{\text{On}} b_i$, $e_i^{\text{On}} = \exp(-U_i^{\text{On}})$, $D_i^{\text{On}} = 1 + e_i^{\text{On}}$, and $\varphi_i^{\text{On}} = P_i^{\text{On}}(\boldsymbol{b}) = 1/D_i^{\text{On}}$. The exact exponential value satisfies Eq. (EC.36), the secant upper bound in Eq. (EC.37), and its box bounds, while $1 = \varphi_i^{\text{On}} D_i^{\text{On}}$ satisfies every McCormick inequality in Eqs. (EC.41)–(EC.44). Hence, the latent-driver activation outer constraints contain the exact binary logit response.

Finally, set $q_i = \sum_{j \in \mathcal{Z}} x_{ij}$ and $w_i = b_i q_i$. The point (b_i, q_i, w_i) lies on the original product graph over $[0, b^{\max}] \times [0, \bar{q}_i]$ and therefore satisfies the payment McCormick envelope. Under these assignments, the relaxed supply constraint coincides with the original supply constraint, and the relaxed objective and budget coincide with their original counterparts. Thus, every feasible DSMDP solution has a feasible lift into $\overline{\text{DSMDP}}$ with the same objective value, which gives $Z^\star \le \overline{Z}$. □

EC.5.7. Solver Bound and Feasible Recovery

We report the solver-certified upper bound U^{solver} only when a valid bound is available after an optimal or time-limit termination. It satisfies $Z^\star \le \overline{Z} \le U^{\text{solver}}$. If the relaxation is solved to global optimality, then $U^{\text{solver}} = \overline{Z}$.

To construct the feasible Conic benchmark, let $\boldsymbol{b}^{\text{raw}}$ and $\boldsymbol{x}^{\text{raw}}$ denote the bonus and matching variables from the relaxation incumbent, and define $q_i^{\text{raw}} = \sum_{j \in \mathcal{Z}} x_{ij}^{\text{raw}}$. We project $\boldsymbol{b}^{\text{raw}}$ onto $\mathcal{B}$ using outflow-based weights q_i^{raw} when $q_i^{\text{raw}} > 10^{-9}$ and unit weights otherwise. We then recompute driver responses at the projected bonus vector and solve DSMDP-M, which enforces the realized bonus-spending constraint $\sum_{i,j \in \mathcal{Z}} b_i x_{ij} \le B$. The resulting solution is feasible for the original DSMDP, and its net operating profit Z^{Conic} is reported separately from U^{solver}. Thus, whenever U^{solver} is available, $Z^{\text{Conic}} \le Z^\star \le \overline{Z} \le U^{\text{solver}}$.

EC.6. Experimental Settings and Benchmarks

All feasible methods are compared under a common protocol, while the solver-certified upper bound is reported separately. We specify the shared data generation, initialization, randomization, solver, timing, and ISA settings, then define the benchmark implementations and the within-protocol fixed-time and time-to-target comparisons. The resulting protocol separates feasible method performance from the quality information supplied by the relaxation bound.

EC.6.1. Parameter Settings

We vary the system configurations across experiments while keeping the following parameters fixed. For rider demand generation, the base demand level is $d^{\text{base}} = 5.0$, the demand-spread parameter is $\sigma^d = 2.0$, and the hotspot intensity is $\theta^d = 25.0$. For initial dispatchable-supply generation, the baseline dispatchable-supply level is $s^{\text{On,base}} = 3.0$, the activity-center intensity is $\theta^s = 8.0$, and the supply-spread parameter is $\sigma^s = 1.5$. The supply-shift radius is $r^s = 4.0$, and the supply-shift strength is $\phi^s = 0.005$.

For latent-supply generation, we set the latent-supply-to-demand ratio to $\rho^{\text{Off}} = 0.30$ and the lognormal noise parameter to $\sigma^{\text{Off}} = 0.3$. We generate the resulting supply from a demand-weighted raw latent-supply vector and then rescale it so that total latent supply equals ρ^{Off} times total demand. For pre-bonus matching-value generation, the pickup-distance term uses an exponential distance-decay function with base value $\theta^p = 15.0$ and decay rate $\xi^p = 0.5$. Destination attractiveness uses spread 2.5, intensity 12.0, and weight 0.7. The behavioral-response parameters are set to $\beta^{(1)} = 0.1, \beta^{(3)} = -0.25, \delta_i^{\text{Off}} \equiv 0.03, \zeta^{\text{Off}} = 0.15, \alpha_i^{\text{On}} \equiv -2.0, \gamma^{\text{On}} = 1.5$. Unless the corresponding sensitivity analysis varies one of these values, the generated experiments use $B = 500$, $b^{\max} = 15$, and $\Delta^b = 1$. All generated instances use data seed 42. The behavioral-parameter estimation error experiment uses the three initial dispatchable-supply-to-demand ratios $\kappa \in \{0.50, 1.00, 1.10\}$ and 10 independent parameter draws for each combination of ratio and perturbation level; its draw streams are indexed by the ratio, perturbation level, and draw number.

For ISA, the initial map allocates $0.3B$ across zones in proportion to demand and is projected onto $\mathcal{B}$. The initial sampling radius is $\epsilon_0 = 0.01 b^{\max} = 0.15$, and the initial stationarity scale is $\nu_0 = 0.1 \max\{1, \|\boldsymbol{g}^0\|_2\}$. Both scales contract by a factor of $\kappa_\epsilon = \kappa_\nu = 0.5$. The sample pool contains at most $2|\mathcal{Z}| + 1$ sensitivities, begins with the center and up to eight ambient samples, and adds ambient samples in batches of at most eight. A scale contraction is preceded by a residual test using at least $|\mathcal{Z}| + 1$ ambient sensitivities. The projected bundle Armijo search uses initial step 1, minimum trial step 10^{-14}, step contraction 0.5, Armijo parameter 10^{-4}, and at most 70 backtracking trials. The main comparisons use sampling seed 2026. Each ordinary ISA phase has a 120-second optimization deadline, 300 accepted steps, and 1,500 sampling attempts; the extended New York

City restart has an additional 1,800-second optimization deadline with the same accepted-step and attempt limits. NOMAD has a 120-second optimization deadline and 2,000 black-box evaluations, and the conic outer relaxation has a 1,800-second optimization deadline. Unless stated otherwise, the target sampling and stationarity scales are one third of their initial values, and the target objective-stability test uses a window of 20 attempts and a relative threshold of 10^{-4}. Every ISA execution returns the best center reached before its stopping condition or computation limit.

EC.6.2. Anytime Performance and Efficiency of the Matching LP

Table EC.1 provides detailed results for the 26 unique generated instances using the comparison measures defined in Section 6.6. We record each improving feasible incumbent and the cumulative number of matching-LP solves, and summarize the distributions using the median and interquartile range (IQR). Panels A–C report ISA's objective lead over NOMAD at fixed times, attainment of the common targets, and ISA's objective lead over the best completed non-ISA feasible solution, respectively.

Table EC.1 Anytime performance on 26 unique generated instances.

Panel A: Fixed-time ISA objective lead over NOMAD

Time (s)	Wins	Losses	Median (%)	IQR (%)	Range (%)
10	26	0	14.36	[11.62, 22.00]	[0.02, 33.84]
30	25	1	13.06	[4.79, 19.84]	[−0.26, 32.62]
60	26	0	11.31	[5.33, 16.47]	[0.36, 32.62]
120	26	0	7.19	[4.85, 13.42]	[0.36, 18.74]

Panel B: Attainment of common within-protocol targets

α	ISA instances	NOMAD instances	ISA time (s) median [IQR]	Max. time (s)	ISA LP solves median [IQR]
0.90	26	0	2.72 [1.26, 17.16]	89.84	132.00 [54.00, 301.25]
0.95	26	0	6.36 [1.41, 18.08]	93.87	209.50 [114.75, 318.25]
0.99	26	0	14.61 [4.57, 36.31]	117.44	468.50 [315.00, 640.75]

Panel C: ISA lead over the best completed non-ISA feasible solution

ISA time (s)	Wins	Losses	Median (%)	IQR (%)	Range (%)
30	25	1	5.62	[3.45, 9.49]	[−0.26, 18.53]
60	26	0	5.94	[4.85, 12.73]	[0.36, 18.74]

ISA attains all three common targets on every instance within the operating window. As the target increases from 90% to 99%, the median attainment time increases from 2.72 to 14.61 seconds, and the median number of matching-LP solves increases from 132 to 468.5. At the 95% target, ISA requires a median of 209.5 matching-LP solves. Its 30-second incumbent exceeds the best completed non-ISA feasible solution on 25 of 26 instances; by 60 seconds, it does so on all 26, with a median objective improvement of 5.94%.

EC.6.3. Benchmark Implementations

Baseline 1: MNL-Response-Aware Greedy Heuristic. $\mathbf{H}_1$ constructs a one-pass driver surge map from the no-bonus map using the score

$$\psi_m = \underbrace{\max\{d_m - s_m(\mathbf{0}), 0\}}_{\text{demand shortage}} \underbrace{\max_{j\in\mathcal{Z}} p_{mj}}_{\text{matching value}} \underbrace{\sum_{i:\, m\in\mathcal{Z}_i} s_i^{\mathrm{On}} P_{im}(\mathbf{0})\big(1 - P_{im}(\mathbf{0})\big)}_{\text{MNL responsiveness}}. \tag{EC.55}$$

After solving the no-bonus matching LP for $\boldsymbol{x}^{\mathrm{no}}$, $\mathbf{H}_1$ uses the score in Eq. (EC.55) to set $\hat{b}_m = \max\{\psi_m, 0\}$ and, when the denominator is positive, rescales $\hat{\boldsymbol{b}} \leftarrow B\hat{\boldsymbol{b}}/(\sum_{m\in\mathcal{Z}} \hat{b}_m \sum_{j\in\mathcal{Z}} x^{\mathrm{no}}_{mj})$. It then clips the bonus vector at $b^{\max}$, projects it onto $\mathcal{B}$, and re-solves DSMDP-M to obtain $(\boldsymbol{b}^{\mathbf{H}_1}, \boldsymbol{x}^{\mathbf{H}_1}, Z^{\mathbf{H}_1})$.

Baseline 2: LP-Dual-Informed Heuristic. The second heuristic, denoted by $\mathbf{H}_2$, uses a selected optimal branch of the no-bonus matching LP. Let $\boldsymbol{x}^{\mathrm{no}}$, $\boldsymbol{\eta}^{\mathrm{no}}$, and λ^{no} denote its primal solution, supply duals, and budget dual, respectively. Define

$$\begin{aligned} q^{\mathrm{no}}_m &:= \sum_j x^{\mathrm{no}}_{mj}, \\ g^{\mathrm{no}}_m &:= \sum_{i\in\mathcal{Z}} \eta^{\mathrm{no}}_i \frac{\partial s_i(\mathbf{0})}{\partial b_m} - (1+\lambda^{\mathrm{no}}) q^{\mathrm{no}}_m, \\ \bar{g} &:= \sum_{m\in\mathcal{Z}} g^{\mathrm{no}}_m. \end{aligned} \tag{EC.56}$$

Using the quantities defined in Eq. (EC.56), $\mathbf{H}_2$ returns $\boldsymbol{b}^{\mathbf{H}_2} = \mathbf{0}$ if $\bar{g} < 0$ or $\max_i \eta^{\mathrm{no}}_i = 0$. Otherwise, for tuning parameters $c_\tau > 0$, $\tau^{\max} > 0$, and $\omega \in [0,1]$, it sets $\tau = \min\{\bar{g}/c_\tau, \tau^{\max}\}$ and $\hat{b}_i = \tau[(1-\omega) + \omega\eta^{\mathrm{no}}_i / \max_\ell \eta^{\mathrm{no}}_\ell]$. It then clips the bonus vector to $[0, b^{\max}]$, adjusts neighboring bonus differences, and clips the vector again. Finally, it recomputes driver responses and re-solves DSMDP-M under the resulting $\boldsymbol{b}^{\mathbf{H}_2}$ for $\boldsymbol{x}^{\mathbf{H}_2}$ and $Z^{\mathbf{H}_2}$.

Baseline 3: NOMAD. NOMAD applies mesh adaptive direct search (Le Digabel 2011) to the bonus vector and solves the matching LP for each candidate. The black-box procedure enforces the constraints defining $\mathcal{B}$, while the matching LP enforces the realized bonus-spending constraint. The separation allows NOMAD to search over implementable bonus maps while each black-box evaluation returns the conditional matching optimum.